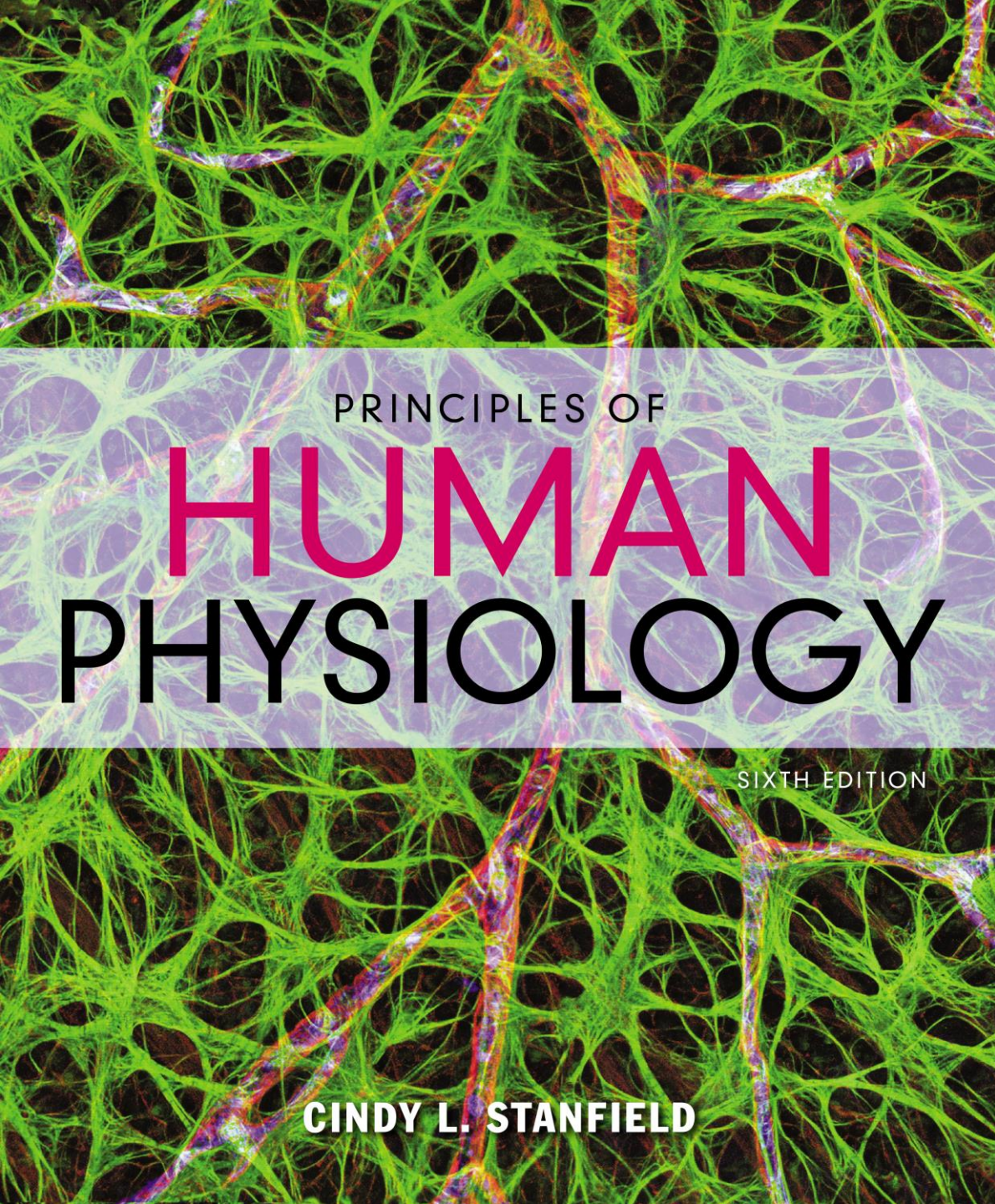




PowerPoint® Lecture
Presentation

PRINCIPLES OF
HUMAN
PHYSIOLOGY

SIXTH EDITION

CINDY L. STANFIELD

CHAPTER **20b**

The
Gastrointestinal
System

Chapter Outline

- 20.1 Overview of Gastrointestinal System Processes
- 20.2 Functional Anatomy of the Gastrointestinal System
- 20.3 Digestion and Absorption of Nutrients and Water
- 20.4 General Principles of Gastrointestinal Regulation
- 20.5 Gastrointestinal Secretion and Its Regulation
- 20.6 Gastrointestinal Motility and Its Regulation

20.4 General Principles of GI Regulation

- Regulation of GI function is not based on the concept of homeostasis
- GI function is regulated to maximize absorption, regardless of whether nutrients are needed
- Conditions in the lumen of GI tract are regulated to maximize absorption

Neural and Endocrine Pathways of GI Control

- Enteric nervous system
 - Submucosal plexus
 - Myenteric plexus
 - Reflexes mediated through ENS control many GI functions
- CNS contributions to neural control
 - Generally through communication of ANS with enteric nervous system

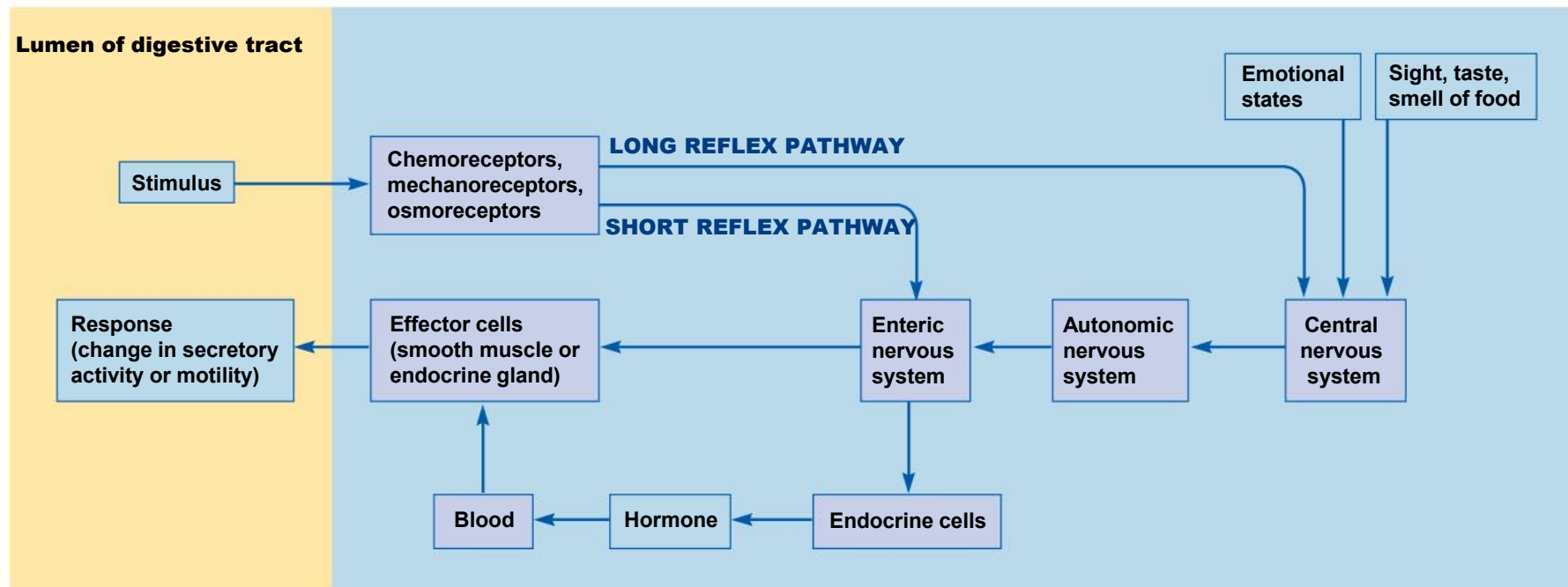
Neural and Endocrine Pathways of GI Control

- GI hormones are secreted from endocrine cells in the stomach and small intestine
 - Cholecystokinin
 - Secretin
 - Glucose-dependent insulinotropic peptide (GIP)
- Sensory receptors in the GI tract detect the environment in the lumen to initiate reflexes

TABLE 20.2 Gastrointestinal Hormones and Their Actions

Hormone	Site of secretion	Stimuli for secretion	Actions
Gastrin	Stomach	Proteins and protein digestion products in stomach; distension of stomach; parasympathetic input to stomach	Stimulates gastric secretion and motility; stimulates ileal motility and relaxes ileocecal sphincter; stimulates mass movement of colon
Cholecystokinin (CCK)	Duodenum and jejunum	Fat or protein digestion products in duodenum	Inhibits gastric secretion and motility; potentiates actions of secretin on pancreatic bicarbonate secretion; stimulates pancreatic enzyme secretion; stimulates bile secretion by liver; stimulates gallbladder contraction and relaxation of sphincter of Oddi
Secretin	Duodenum and jejunum	Acid in duodenum	Inhibits gastric secretion and motility; stimulates pancreatic bicarbonate secretion; potentiates actions of CCK on pancreatic enzyme secretion; stimulates bile secretion by liver
Glucose-dependent insulinotropic peptide (GIP)	Duodenum and jejunum	Glucose, fats, or acid in duodenum; distension of duodenum	Inhibits gastric secretion and motility; stimulates insulin secretion by pancreas

Figure 20.22 Summary of neural and endocrine control of gastrointestinal function.

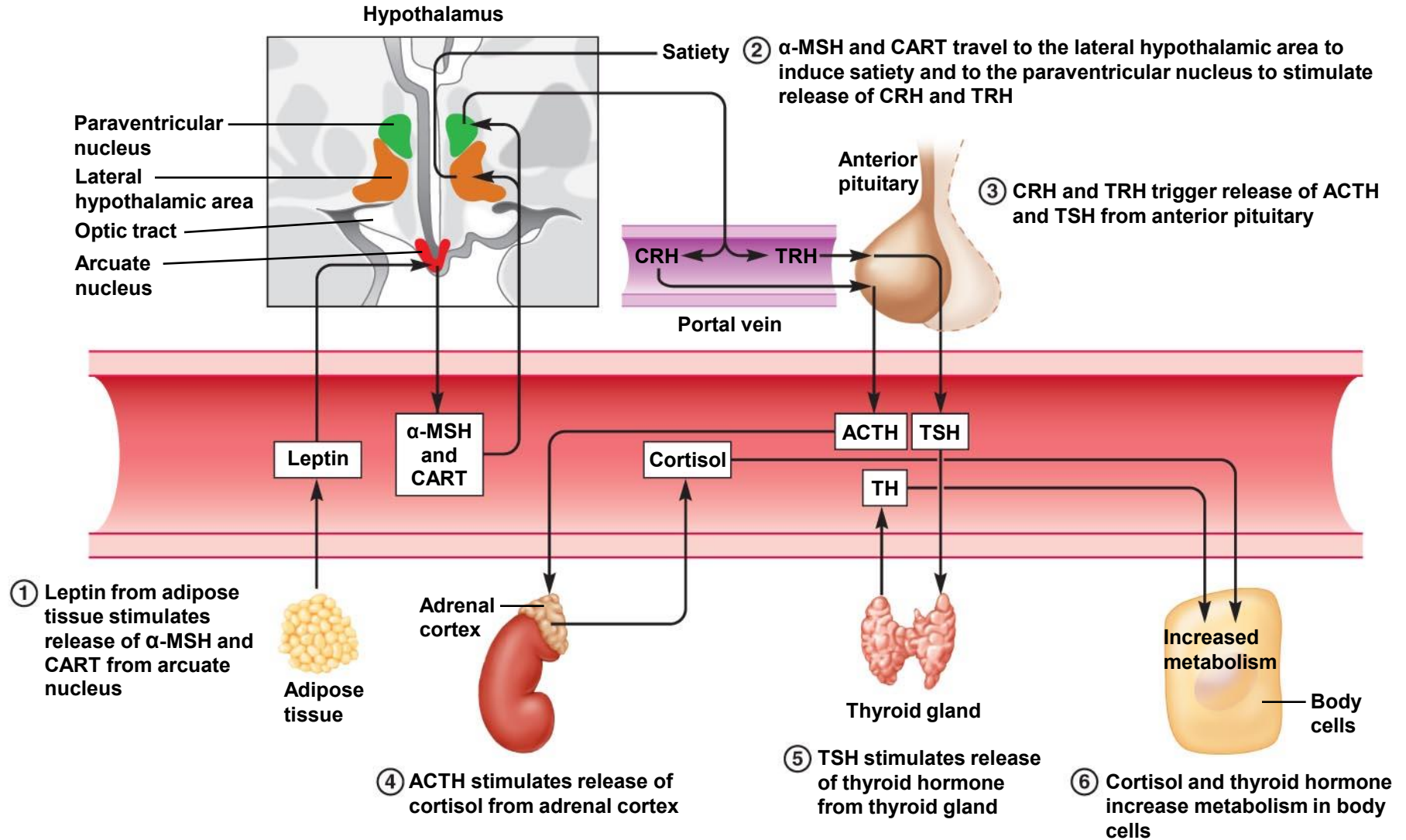


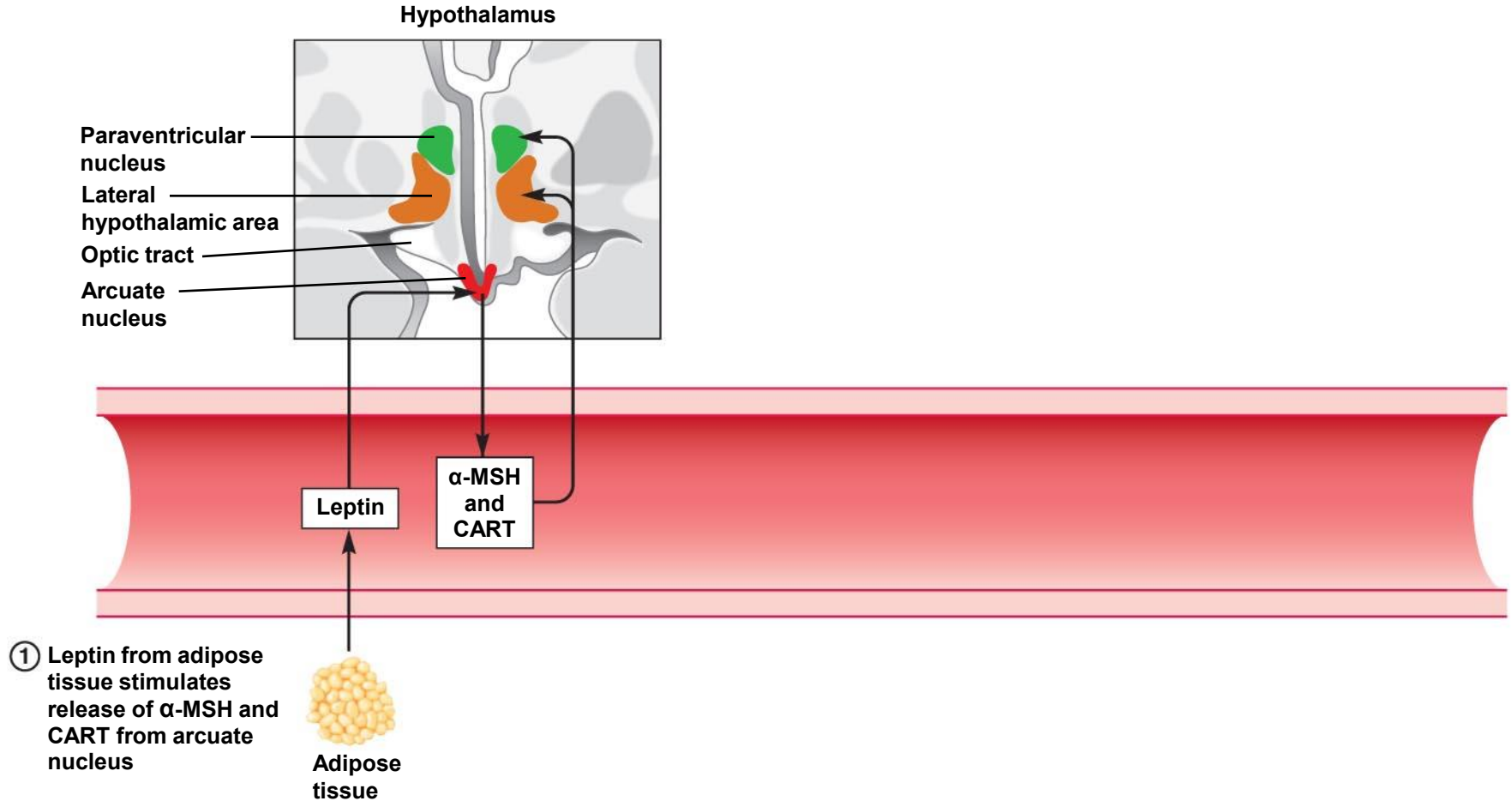
Neural and Endocrine Pathways of GI Control

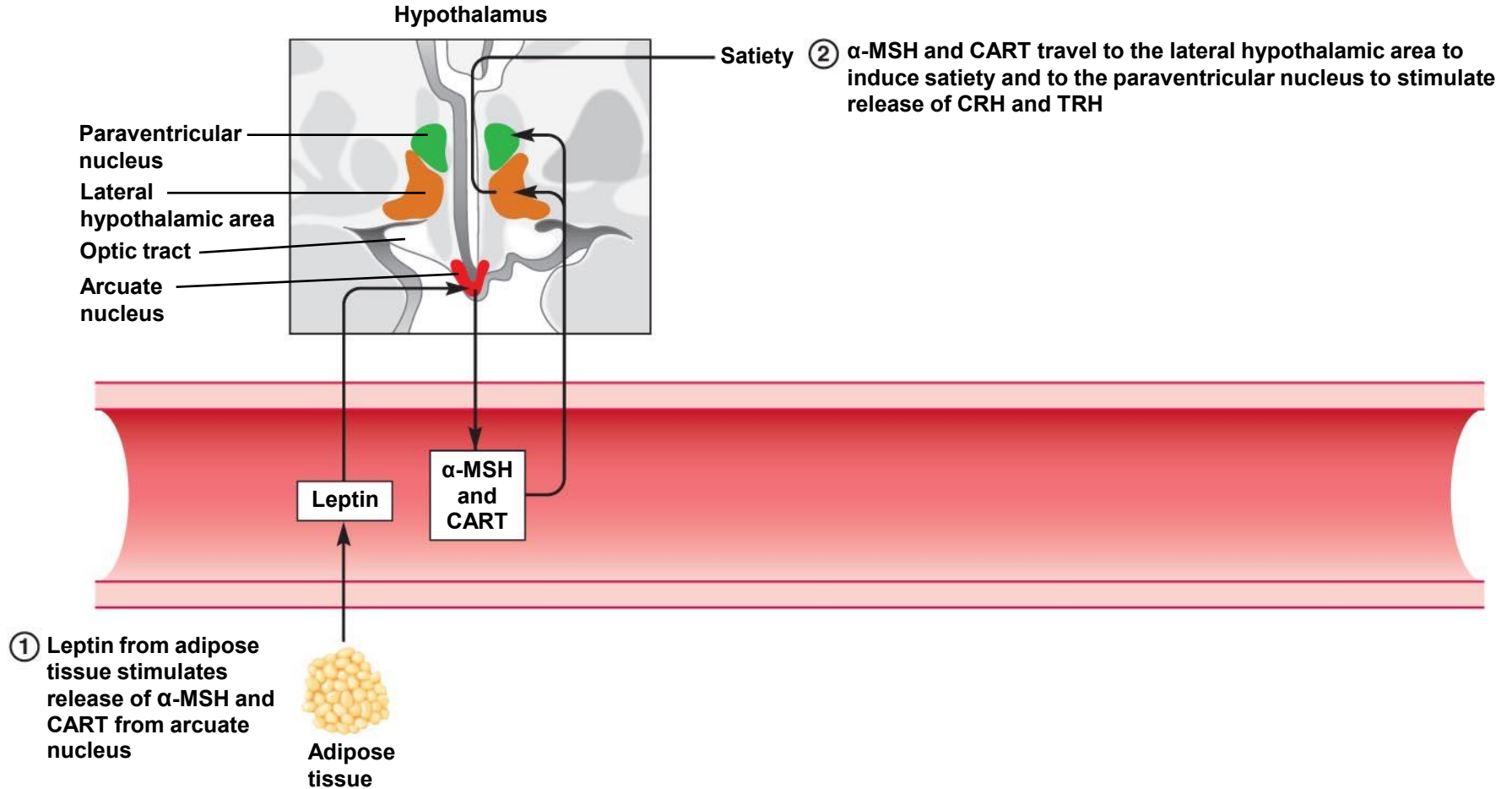
- Phases of gastrointestinal control
 - Cephalic phase: stimuli originate in head
 - Thoughts, taste, and smell
 - Requires input from CNS (long reflexes)
 - Gastric phase: stimuli originate in stomach
 - Long and short reflexes and GI hormones
 - Intestinal phase: stimuli originate in small intestine
 - Long and short reflexes and GI hormones

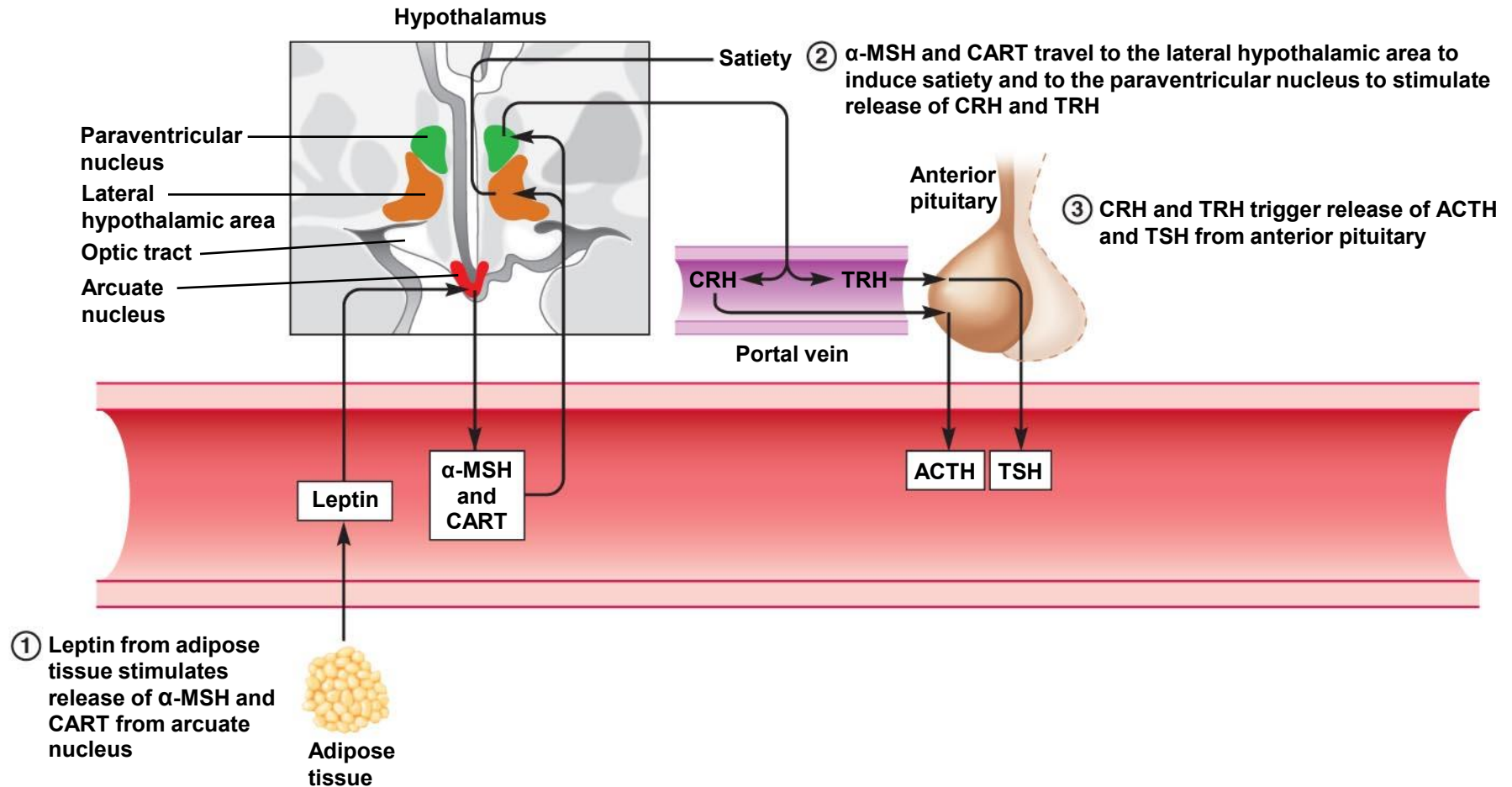
Regulation of Food Intake

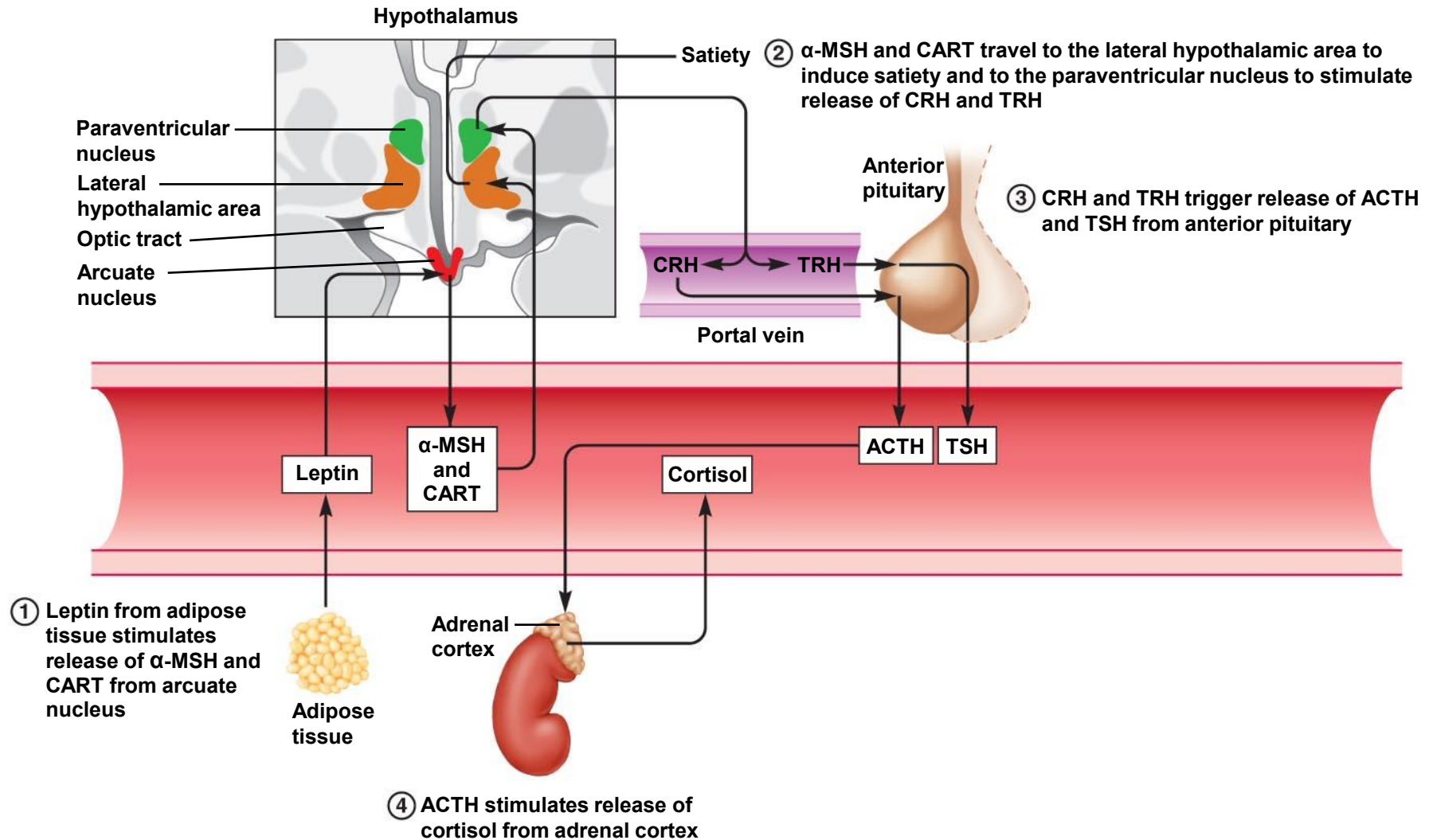
- Leptin: hormone released from adipose cells when calories exceed demands
- Leptin suppresses hunger and increases metabolism: *satiety signal*
- Leptin stimulates α MSH and CART (cocaine- and amphetamine-related transcript)
- α MSH and CART increase sympathetic activity and stimulate release of TSH and ACTH
- TSH and ACTH increase metabolic rate and reduce fat storage

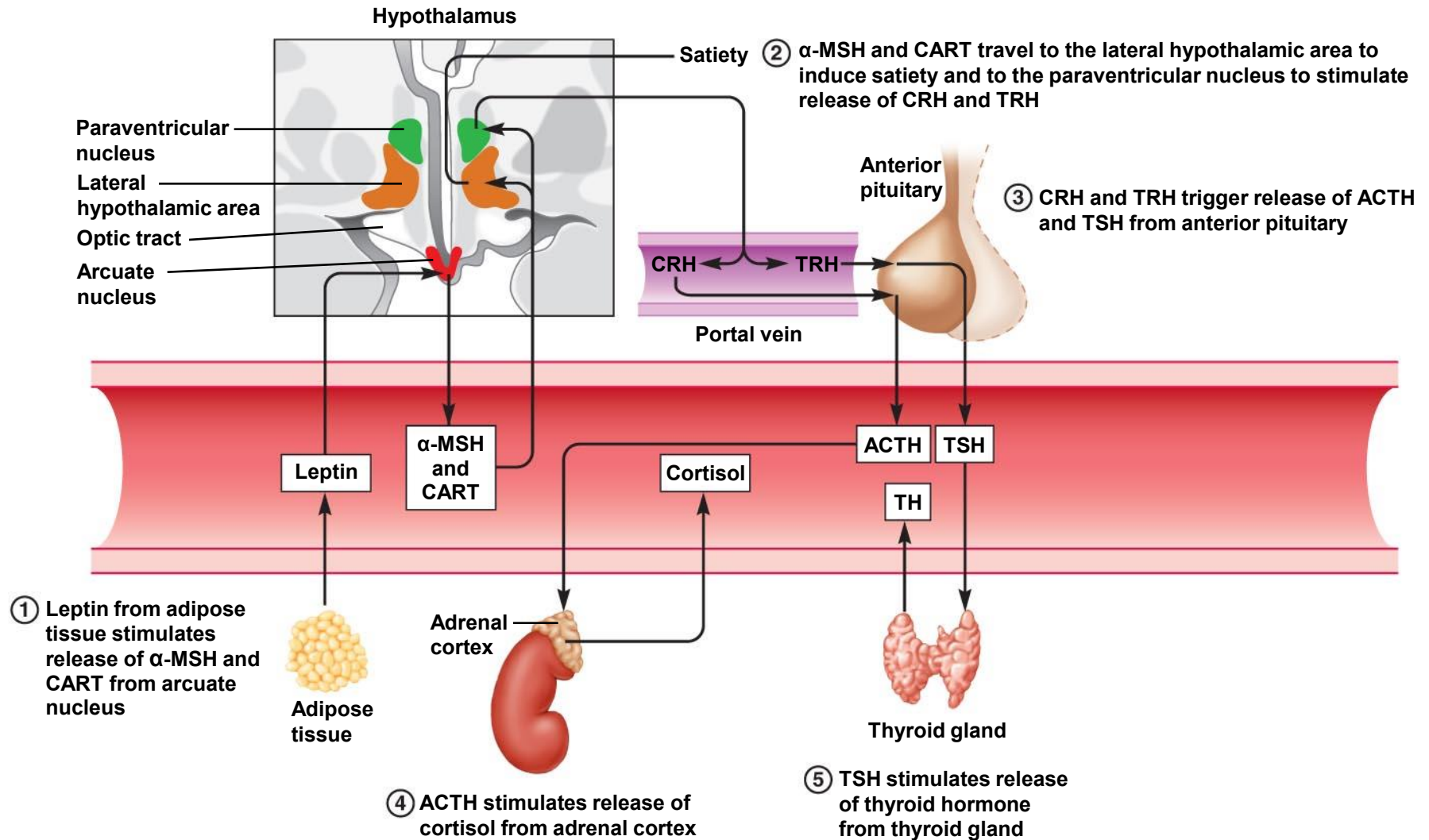


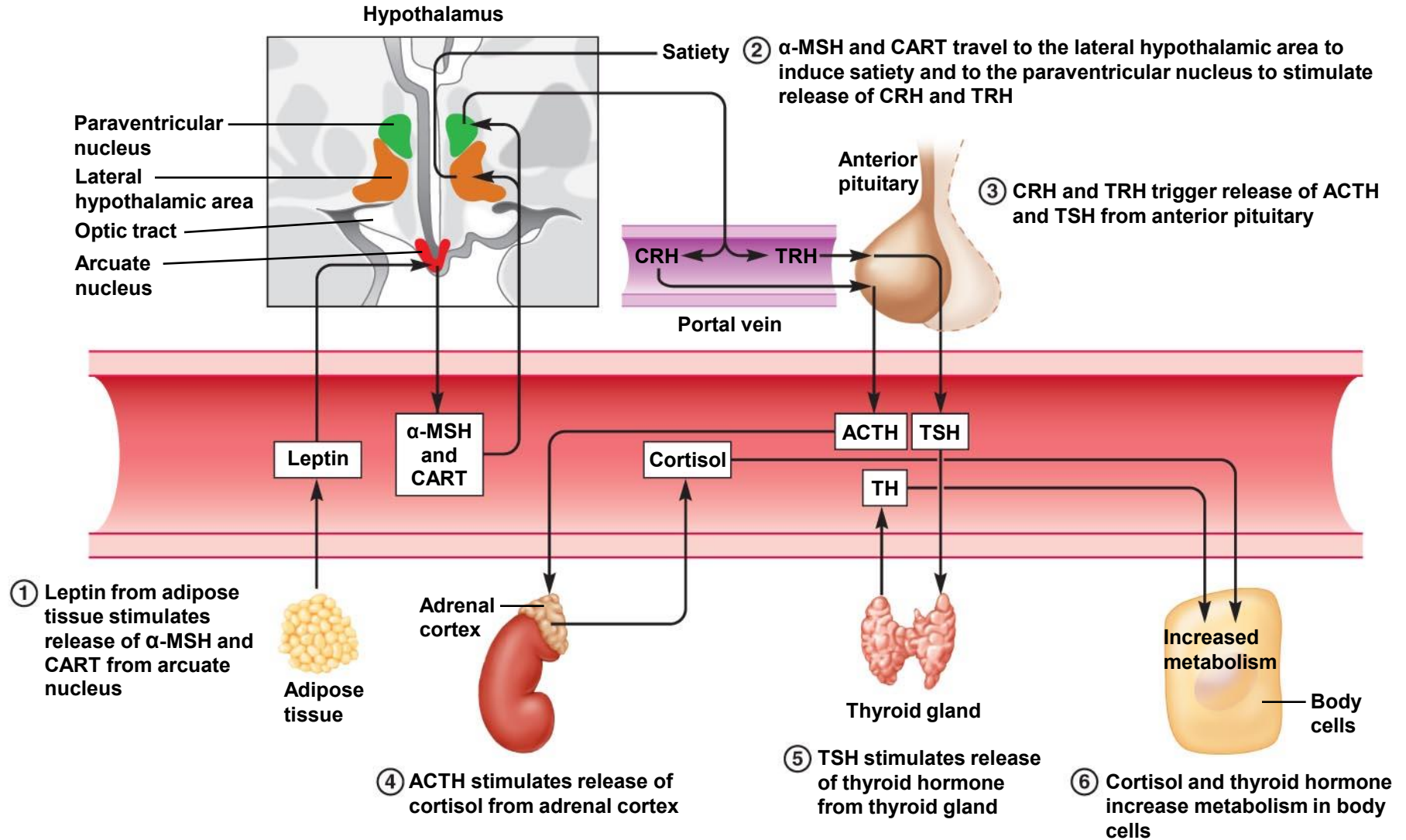












Regulation of Food Intake

- *Orexigenic factors* oppose satiety factors
 - NPY
 - AgRP
- Promote eating
- Decrease metabolism
- Increase parasympathetic activity
- Decrease sympathetic activity
- Inhibit secretion of TSH and ACTH

Regulation of Food Intake

- Short-term regulation: hunger versus satiety
- Satiety factors
 - Insulin
 - CCK
 - Neural input from mechanoreceptors and chemoreceptors
- Orexigenic factors
 - Ghrelin

20.5 Gastrointestinal Secretion and Its Regulation

- Saliva secretion
- Acid and pepsinogen secretion in the stomach
- Secretion of pancreatic juice and bile
- Rates of fluid movement in the digestive tract

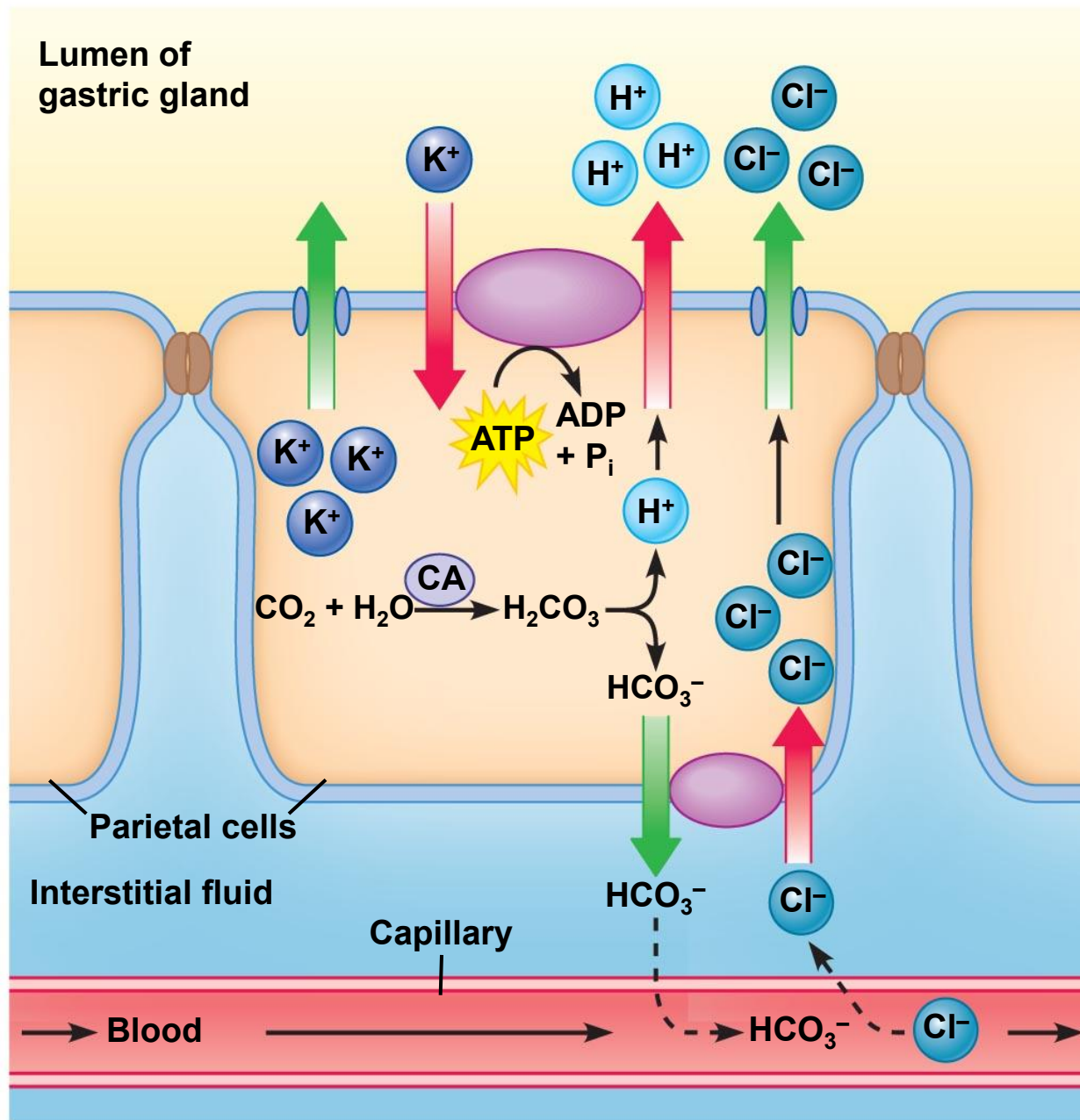
Saliva Secretion

- Taste and texture of food stimulate mechanoreceptors and chemoreceptors in the mouth
- Relayed to the *salivary center* of the medulla, which regulates the autonomic nervous system
- Sight and smell input from cerebral cortex also affects the salivary center
- Salivation stimulated
 - Parasympathetic activation: watery saliva
 - Sympathetic activation: protein-rich, viscous saliva

Acid and Pepsinogen Secretion in the Stomach

- Parietal cells produce hydrochloric acid
 - Carbonic anhydrase catalyzes production of bicarbonate and H^+
 - H^+ is actively secreted into lumen of the stomach
 - Bicarbonate is transported into interstitial fluid for Cl^-
 - Cl^- diffuses into lumen of the stomach through channels
- Net result of acid production
 - H^+ and Cl^- are secreted into lumen of the stomach
 - Bicarbonate enters the interstitial fluid and then blood

Figure 20.24 Mechanism of gastric acid secretion.

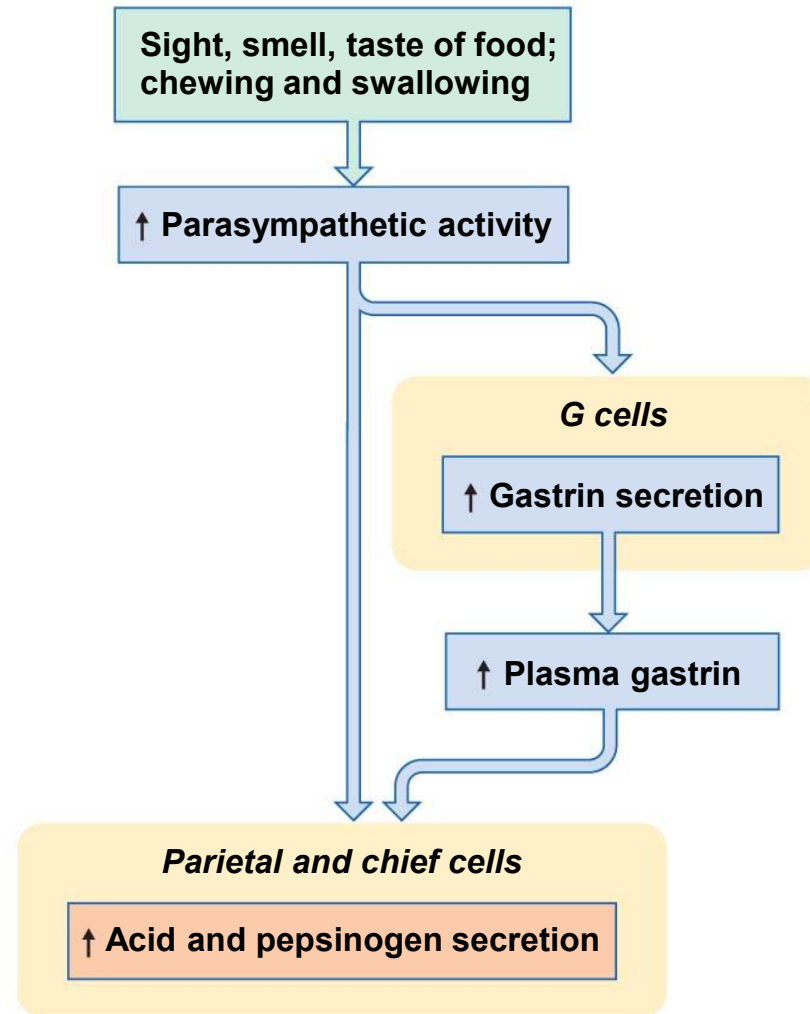


Acid and Pepsinogen Secretion in the Stomach

- Acid secretion
 - Parasympathetic nervous system
 - Gastrin
 - Histamine (paracrine)
- Pepsinogen secretion regulated in sync with acid secretion
 - Parasympathetic nervous system
 - Gastrin
 - Histamine

Acid and Pepsinogen Secretion in the Stomach

- Cephalic-phase regulation of secretion
 - Stimuli increase acid and pepsinogen secretion
 - Sight of food
 - Taste
 - Smell
 - Chewing
 - Swallowing
 - All activate the parasympathetic nervous system
 - Parasympathetic nervous system stimulates gastrin secretion

**(a)** Cephalic-phase control of gastric secretion

- Initial stimulus
- Physiological response
- Result

Sight, smell, taste of food;
chewing and swallowing



(a) Cephalic-phase control of gastric secretion



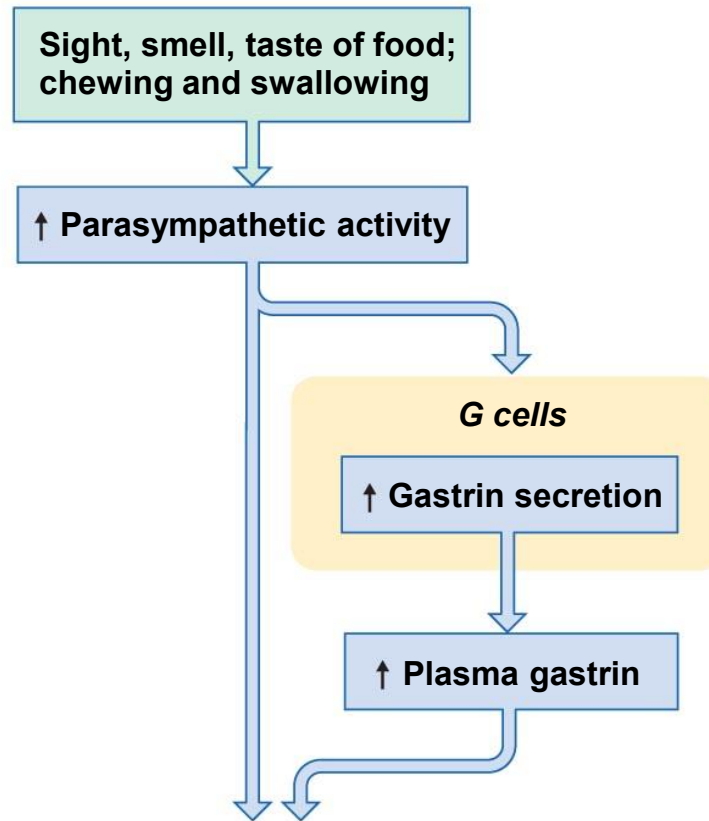
Initial stimulus



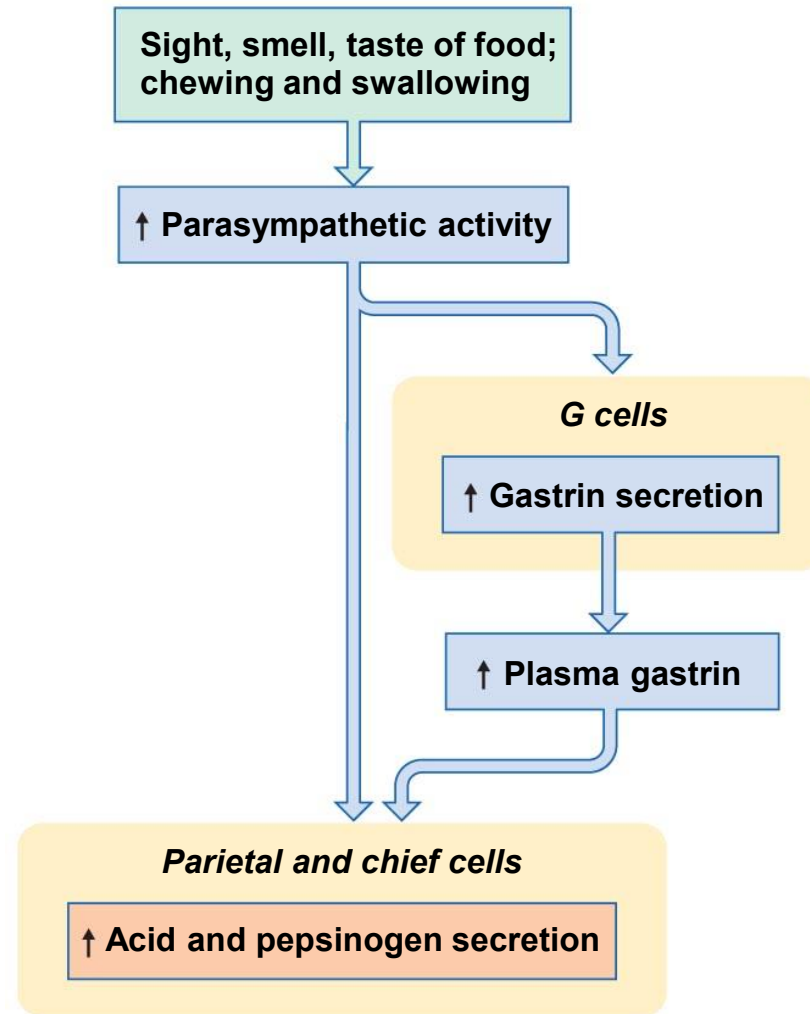
Physiological response



Result

**(a) Cephalic-phase control of gastric secretion**

- Initial stimulus
- Physiological response
- Result

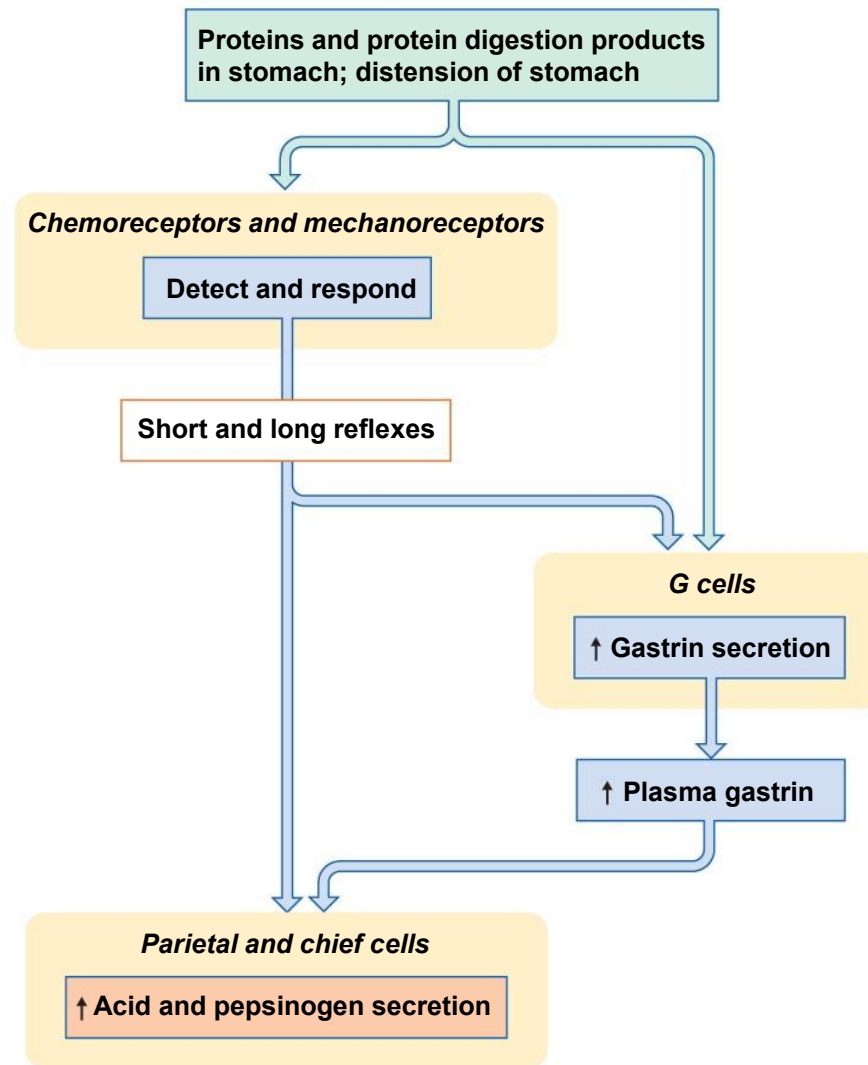
**(a)** Cephalic-phase control of gastric secretion

- Initial stimulus
- Physiological response
- Result

Acid and Pepsinogen Secretion in the Stomach

- Gastric-phase regulation of secretion
 - Stimuli of gastric secretion
 - Proteins, peptides, and amino acids
 - Distension of stomach
 - Short and long reflex pathways trigger gastrin, acid, and pepsinogen release

Figure 20.25b Regulation of gastric secretion.



(b) Gastric-phase control of gastric secretion

- Initial stimulus
- Physiological response
- Result

Acid and Pepsinogen Secretion in the Stomach

- Inhibition of secretion
 - Gastric phase
 - Exit of food removes stimuli for secretion
 - Increased acidity inhibits gastrin release
 - Intestinal phase
 - Effects of food entry into the duodenum
 - Increased osmolarity
 - Increased fat and acid
 - Increased distension
 - Long and short reflex pathways inhibit acid and pepsinogen secretion

Secretion of Pancreatic Juice and Bile

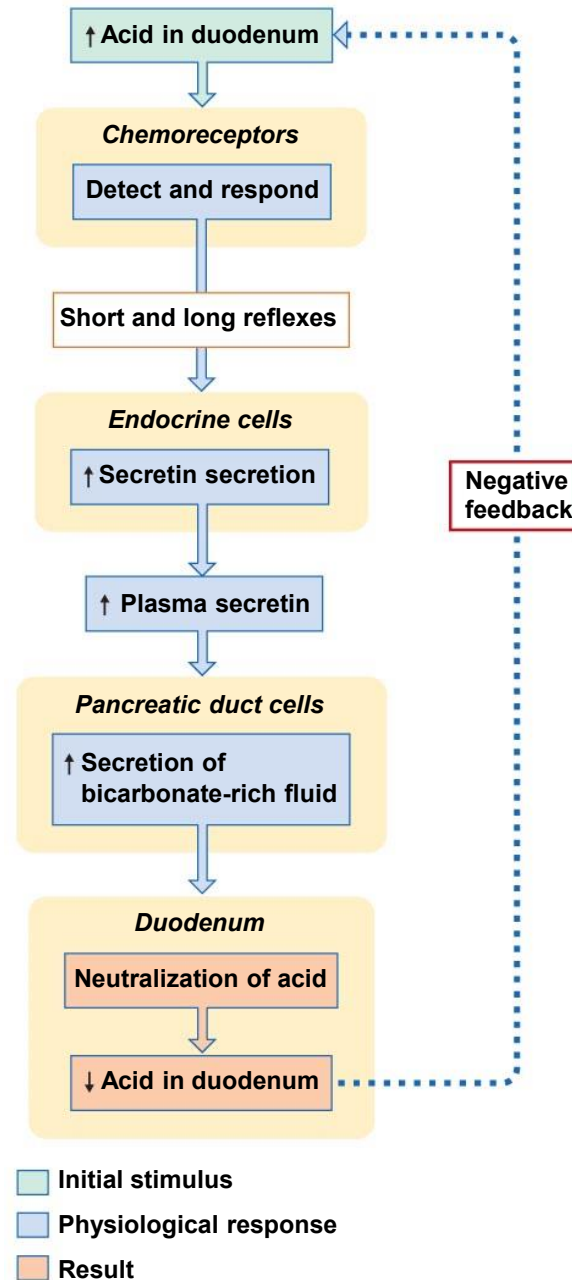
- Acinar cells
 - Small volume of primary secretion
 - Water, electrolytes, and digestive enzymes
- Duct cells
 - Large volume
 - Bicarbonate-rich secretion

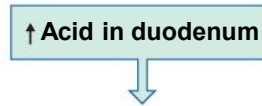
Secretion of Pancreatic Juice and Bile

- CCK and secretin: primary stimulants
 - Enzyme and bicarbonate secretion regulated independently
 - Composition of pancreatic juice varies based on luminal contents
 - CCK stimulates acinar cells to secrete enzymes
 - Small stimulant for bicarbonate release
 - CCK potentiates secretin effects on bicarbonate release
 - Secretin stimulates duct cells to secrete bicarbonate
 - Small stimulant for enzyme release
 - Secretin potentiates CCK effects on enzyme release

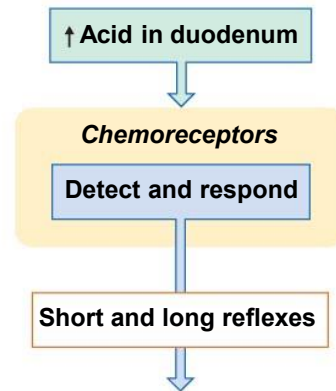
Secretion of Pancreatic Juice and Bile

- Stimuli for CCK release
 - Fat and amino acids in duodenum
- Stimuli for secretin release
 - Acidity in duodenum

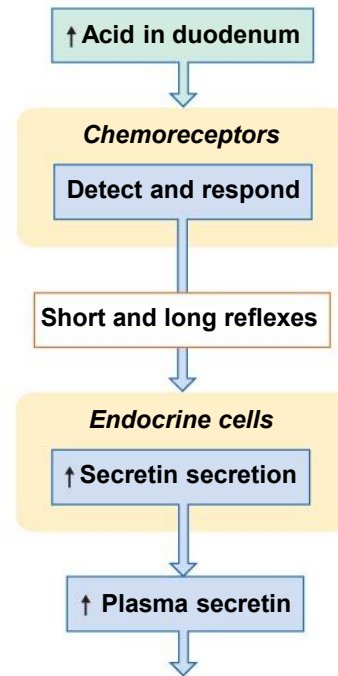




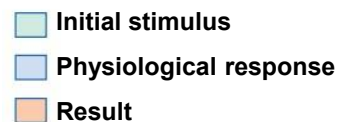
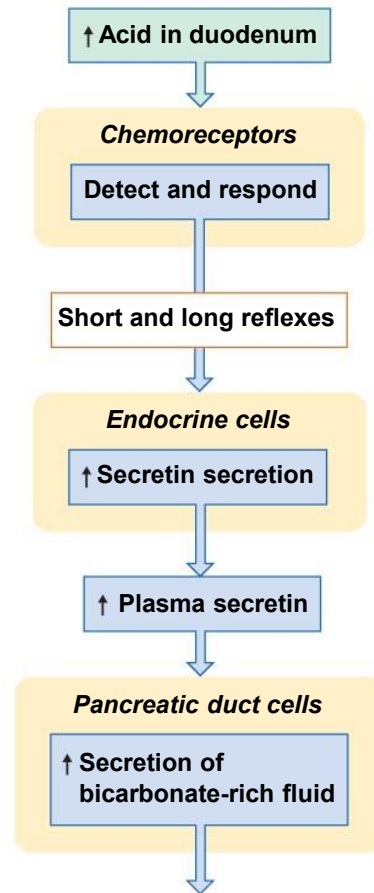
-  Initial stimulus
-  Physiological response
-  Result

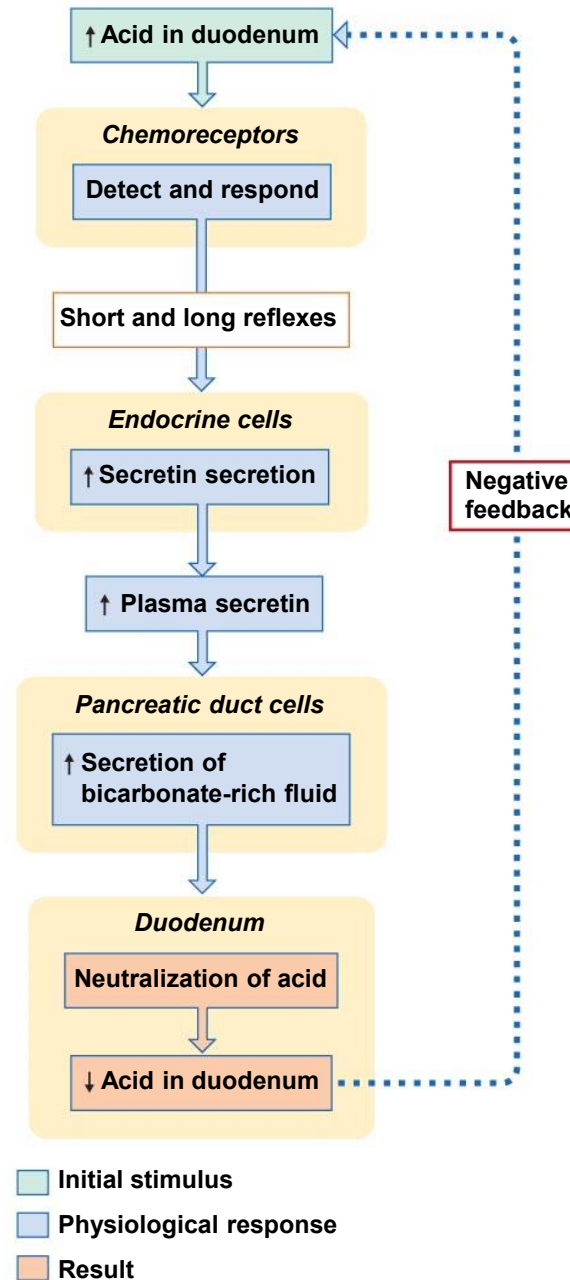


- Initial stimulus
- Physiological response
- Result



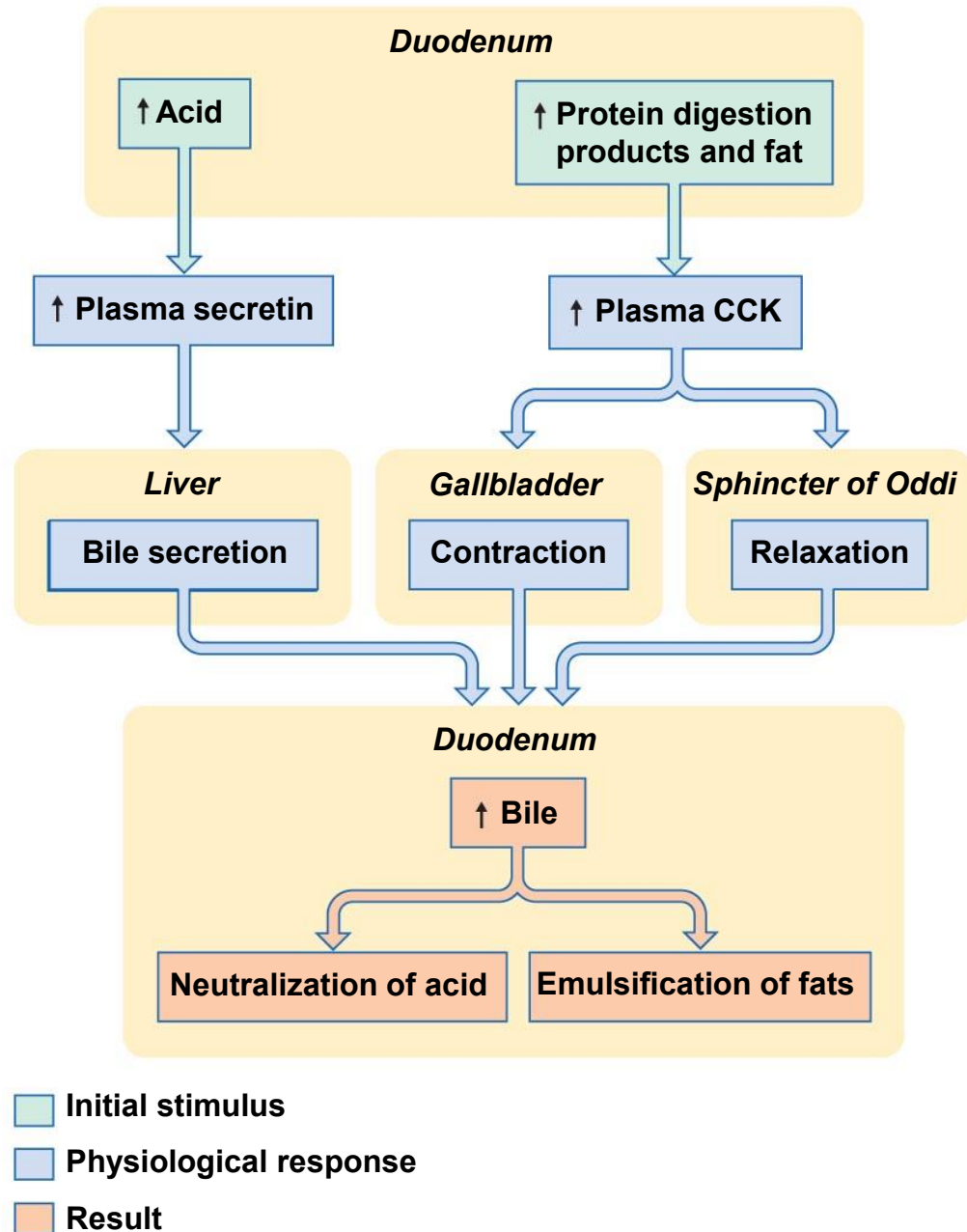
- Initial stimulus
- Physiological response
- Result

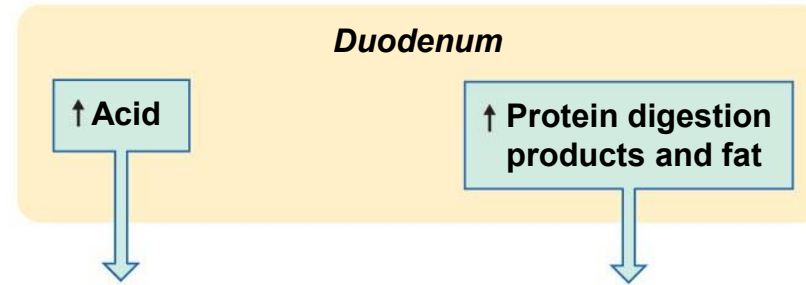




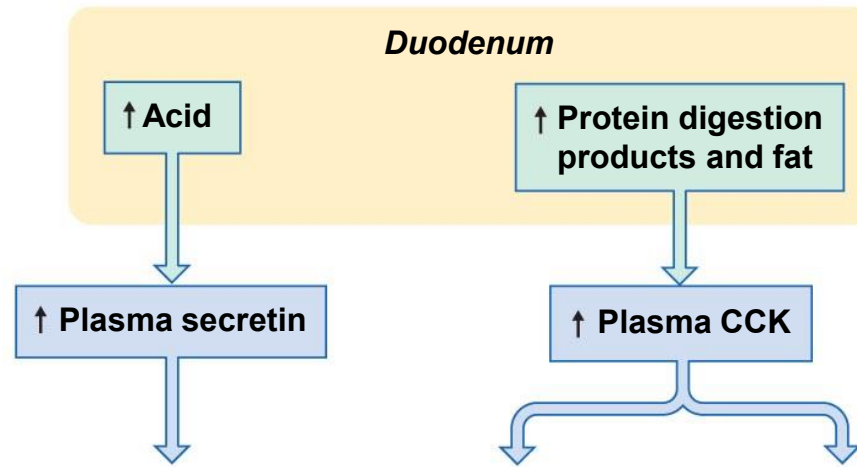
Secretion of Pancreatic Juice and Bile

- Regulation of bile secretion into duodenum
 - Secretin
 - Stimulates bile secretion from liver
 - CCK
 - Stimulates gallbladder contraction
 - Stimulates relaxation of sphincter of Oddi

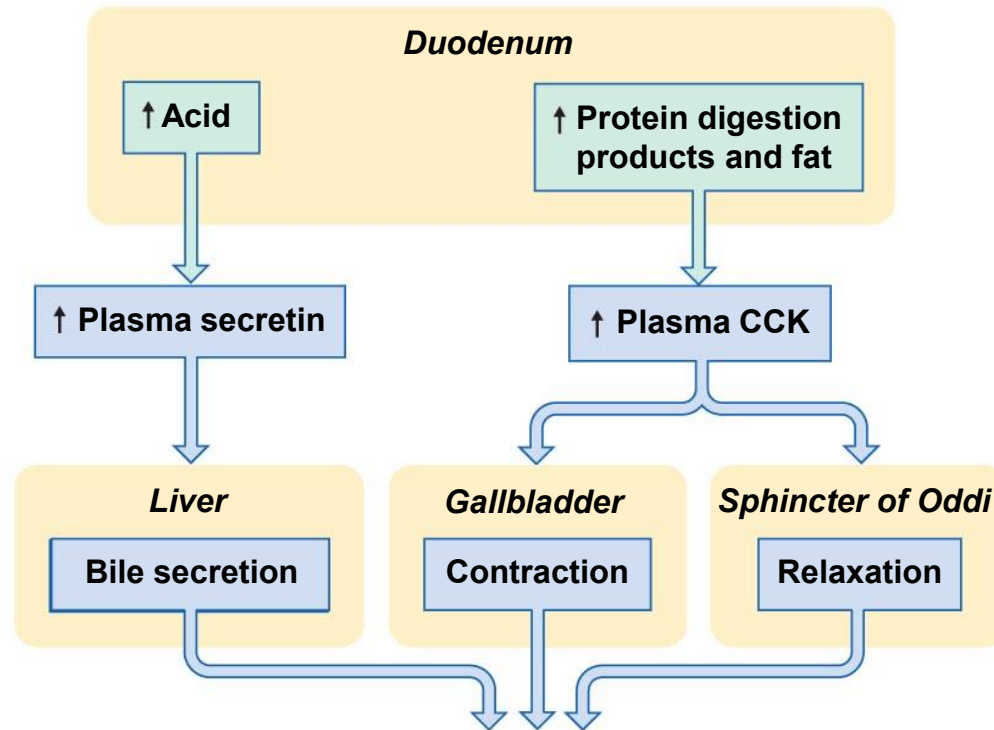




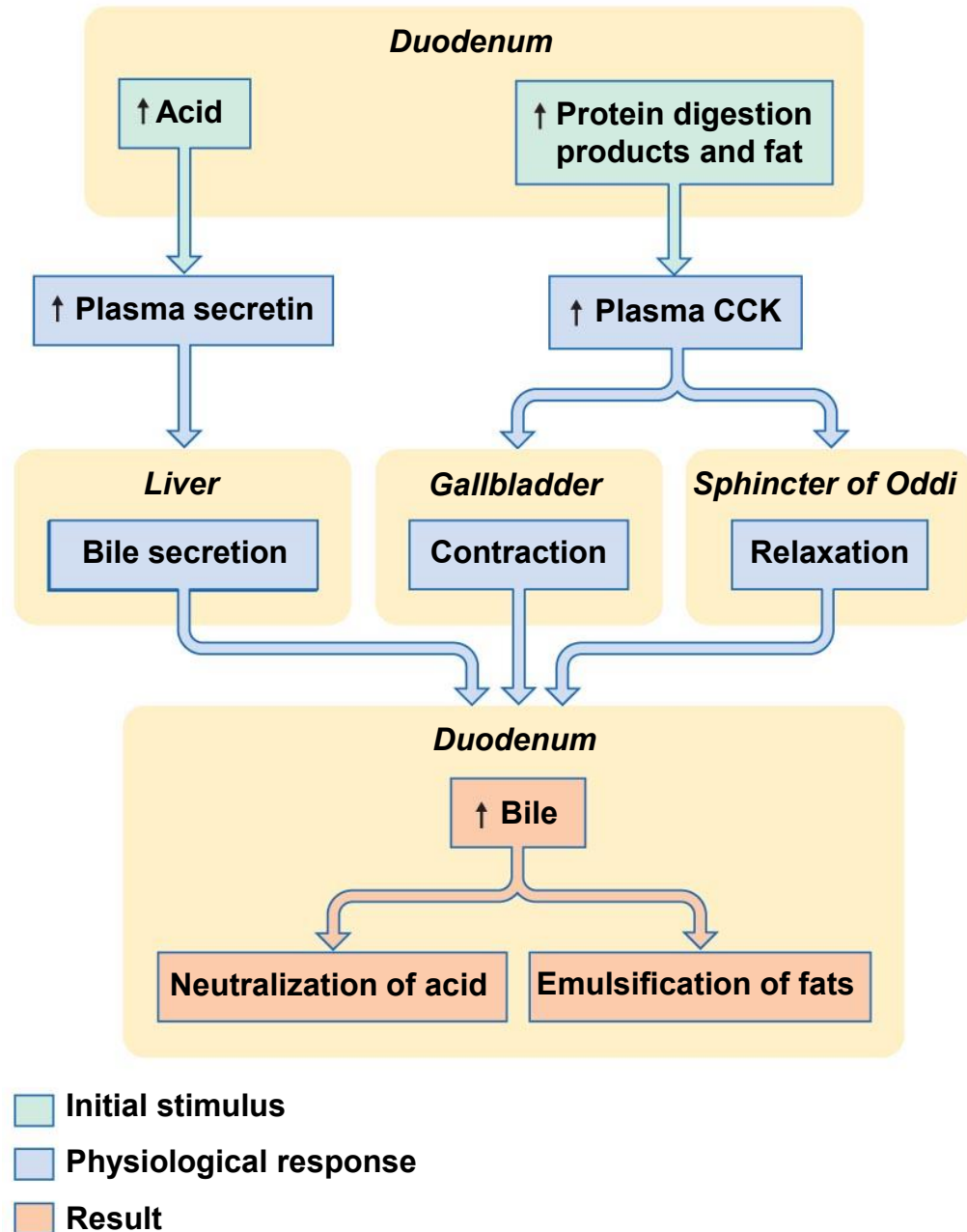
- Initial stimulus
- Physiological response
- Result



- Initial stimulus
- Physiological response
- Result



- Initial stimulus
- Physiological response
- Result



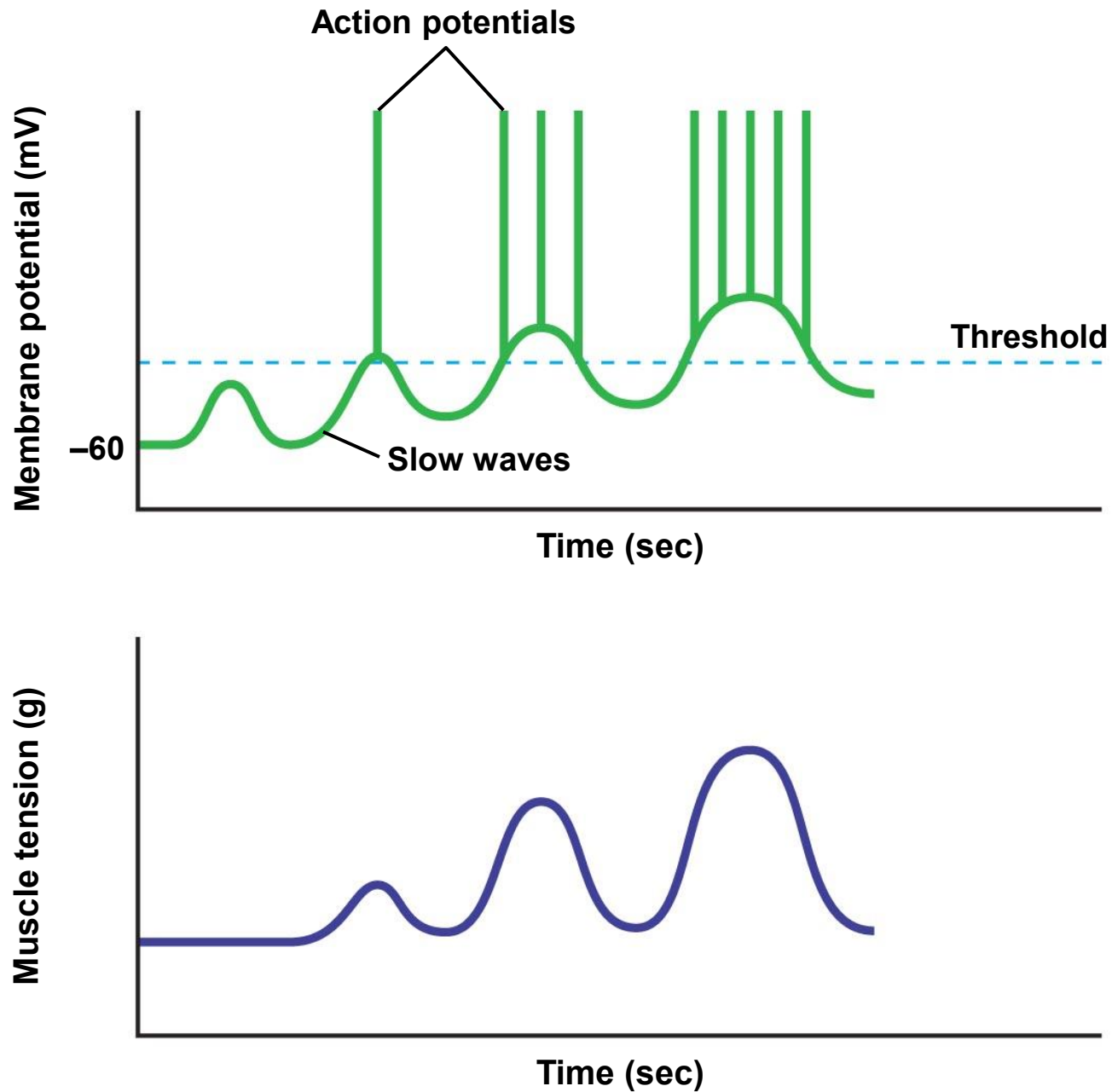
20.7 Gastrointestinal Motility and Its Regulation

- GI motility
 - Movements of the wall of the GI tract
 - Due primarily to contractions of the muscularis externa (outer muscle layers)
 - Function: mix and propel GI tract contents

Electrical Activity in GI Smooth Muscle

- Spontaneous slow waves of depolarization = slow waves
- Frequency of waves = basic electric rhythm
- BER varies in different areas of the GI tract
- Height of BER is affected by neural and hormonal input
 - Parasympathetic → excitation
 - Sympathetic → inhibition

Figure 20.29 Slow-wave electrical activity in gastrointestinal smooth muscle.



Relationship Between Electrical and Mechanical Activity in Smooth Muscle

- Stomach
 - Larger depolarizations → stronger contractions
 - Action potentials → even stronger contractions
- Intestines
 - Action potentials required for force
 - Strength of contraction varies with frequency of action potentials

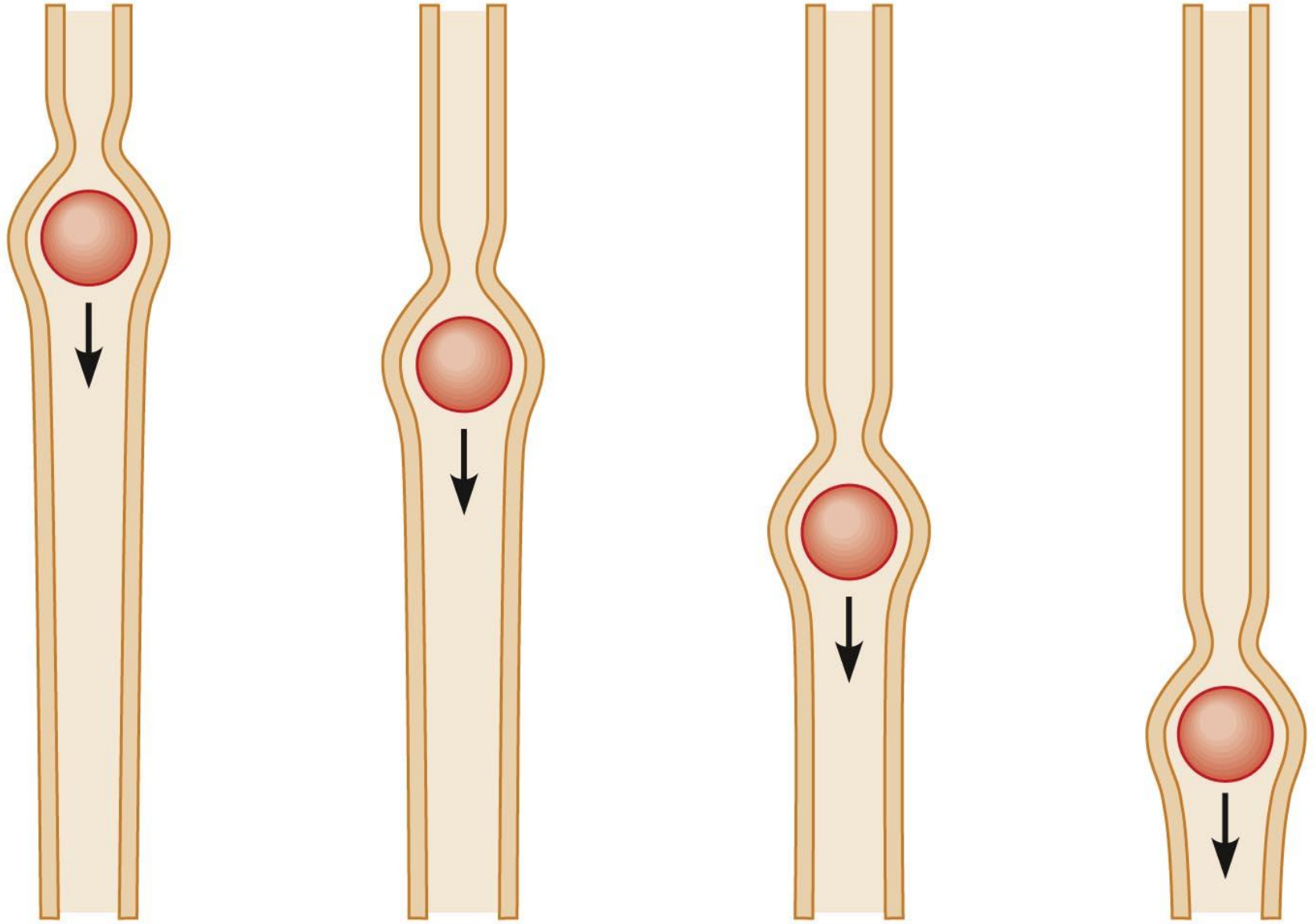
Peristalsis and Segmentation

- Waves of contraction in muscularis mucosae in response to basic electrical rhythm
- Peristalsis
 - Propels contents forward
- Segmentation
 - Mixes contents

Peristalsis and Segmentation

- Peristalsis
 - Proximal segment
 - Circular muscle contracts; longitudinal muscle relaxes
 - Diameter decreases
 - Distal segment
 - Circular muscle relaxes; longitudinal muscle contracts
 - Diameter increases
 - Contents propelled forward from area of small diameter to area of large diameter

Figure 20.30a Comparison of peristalsis and segmentation.

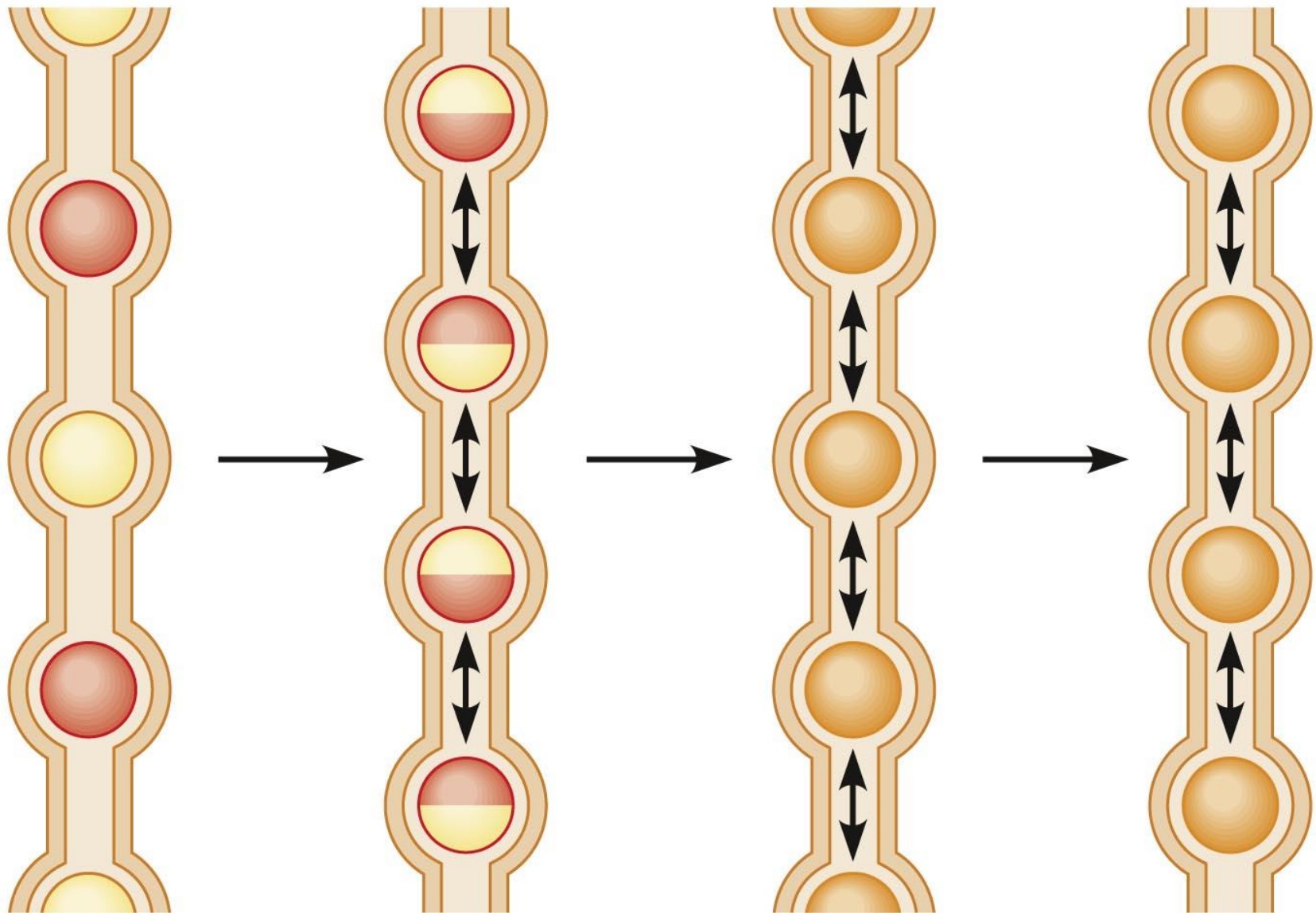


(a) Peristalsis

Peristalsis and Segmentation

- Segmentation
 - A type of motility of the small intestine
 - Requires circular muscle layer
 - Alternating contractions between intestinal segments
 - Mixes chyme

Figure 20.30b Comparison of peristalsis and segmentation.



(b) Segmentation

Chewing and Swallowing

- Chewing
 - Control is both voluntary and involuntary
 - Chewing reflex: cyclical reflex
 - Normally, jaw muscles are active; hold mouth closed
 - Food enters mouth; inhibits jaw muscles
 - Jaw dropping relieves pressure of food; contracts
 - Pressure from food restored; inhibited

Chewing and Swallowing

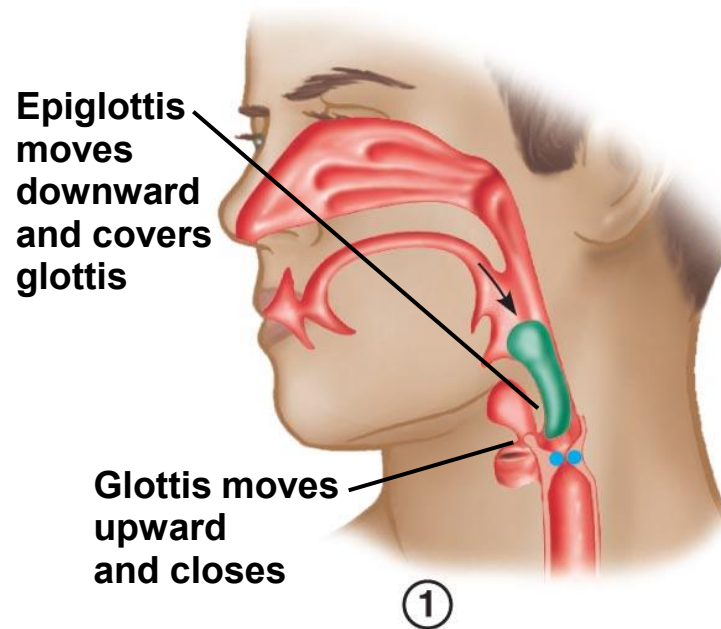
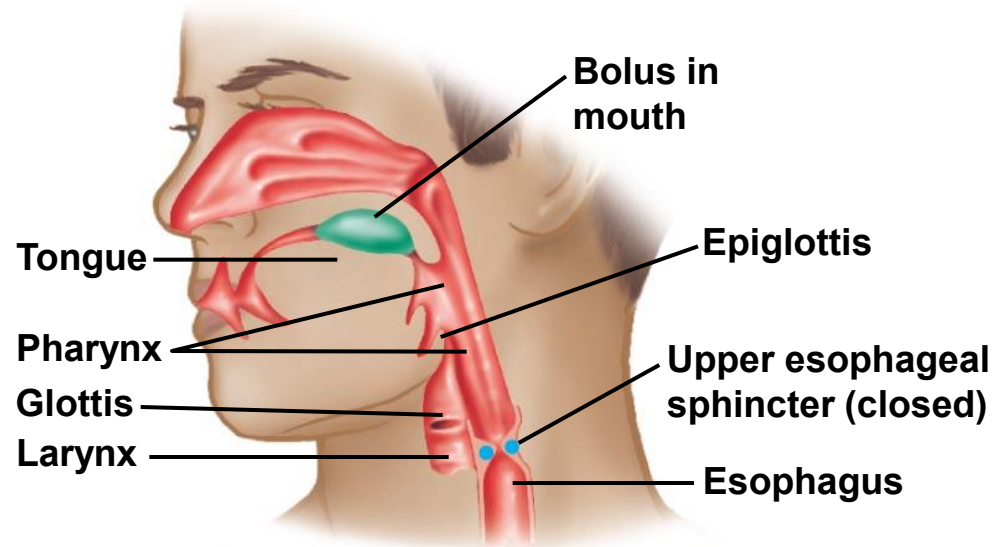
- Swallowing
 - Chewed food + saliva = bolus
 - Tongue moves bolus to pharynx
 - Initiates swallowing reflex
 - Integration center = swallowing center of medulla oblongata

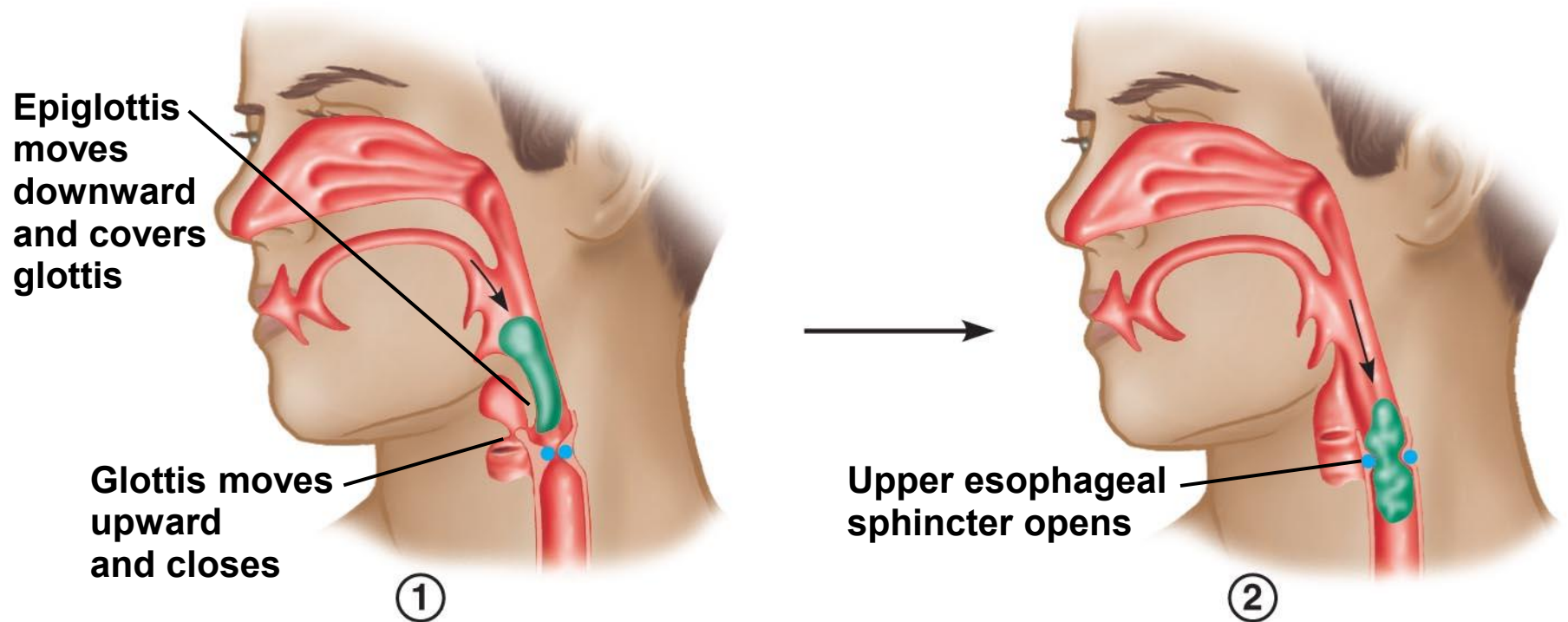
Chewing and Swallowing

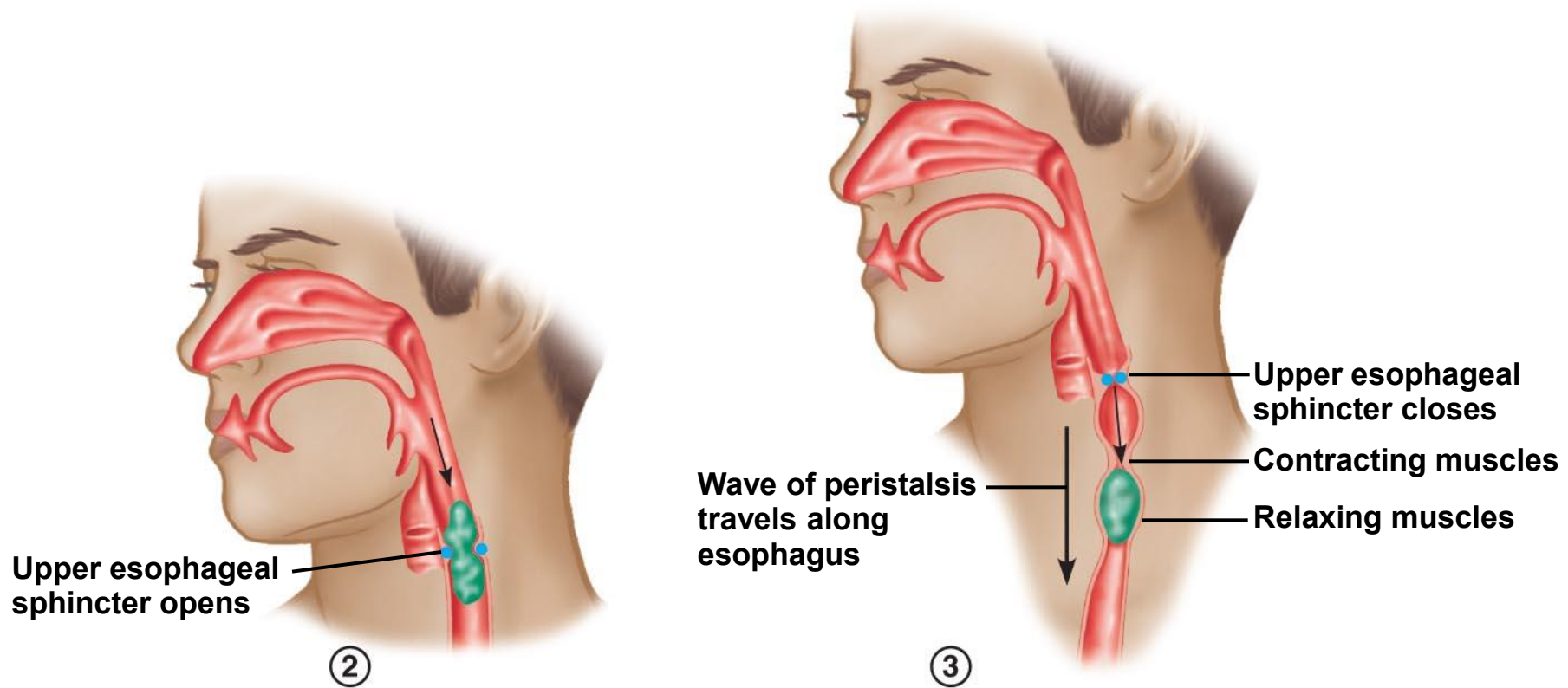
- Swallowing reflex
 - Bolus descends in pharynx, pressing on epiglottis
 - Epiglottis covers glottis so food does not enter trachea
 - Inspiration is inhibited
 - Upper esophageal sphincter relaxes
 - Bolus enters esophagus
 - Sphincter closes behind bolus
 - Bolus stretches esophagus, triggering peristalsis
 - Peristalsis propels bolus to stomach
 - Travel time is approximately 9 seconds

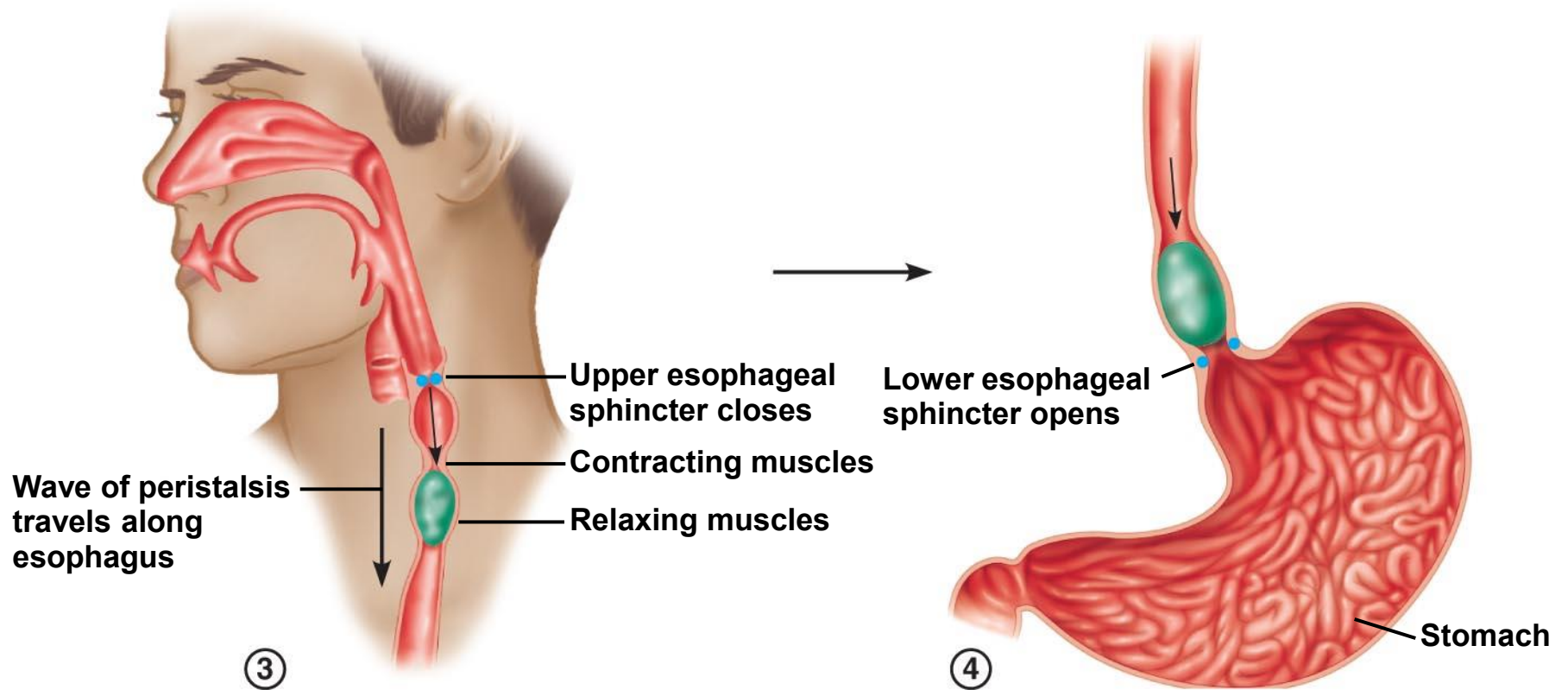
Chewing and Swallowing

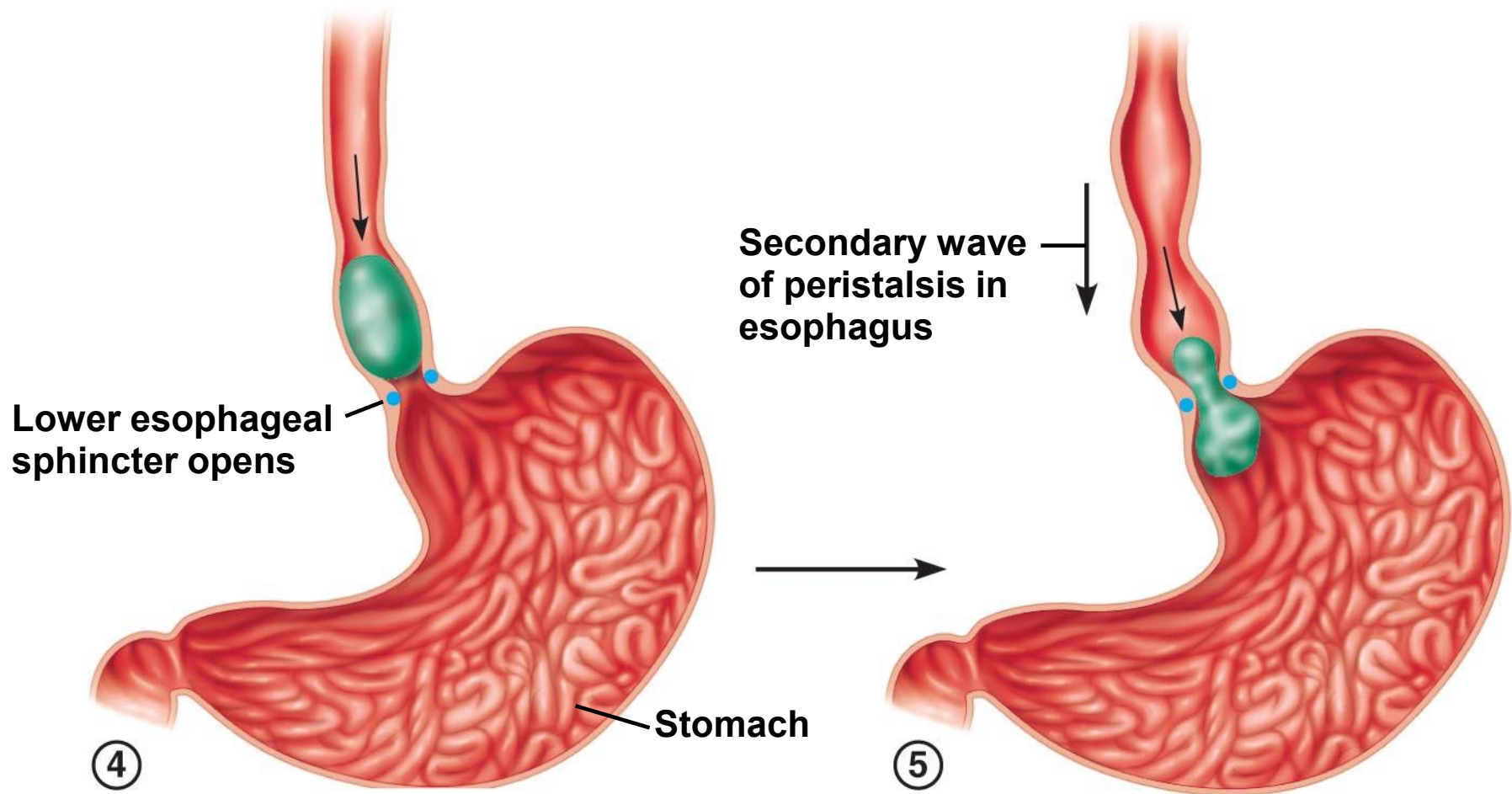
- Swallowing reflex
 - Bolus arrives at stomach
 - Triggers relaxation of lower esophageal sphincter
 - Bolus enters stomach
 - If necessary, secondary peristalsis waves occur to move bolus into stomach
- Receptive relaxation of stomach
 - Swallowing center also initiates relaxation of stomach smooth muscle as it prepares for receipt of bolus











Gastric Motility

- Functions
 - Mix chyme
 - Regulate gastric emptying
- Mechanism
 - Peristalsis
 - Coordinated by enteric nervous system

Gastric Motility

- Gastric motility patterns
 - Waves of peristalsis
 - Upper body → pylorus
 - Strengthen as they approach the pylorus
 - Functions in mixing of chyme
 - Pyloric sphincter closed
 - Functions in gastric emptying
 - Stronger contractions open pyloric sphincter
 - Factors governing emptying rate
 - Volume of chyme in stomach
 - Strength of gastric peristalsis

Gastric Motility

- Migrating motility complex
 - Wave of intense contractions
 - Travels short distances
 - Occurs between meals to clear stomach

Gastric Motility

- Regulation of gastric motility
 - Increase force
 - Gastrin
 - Decrease force
 - CCK
 - Secretin
 - GIP

Gastric Motility

- Phases of regulation
 - Cephalic-phase excitations
 - Anger, aggression
 - Cephalic-phase inhibitions
 - Pain, fear, depression
 - Gastric-phase excitatory stimulus
 - Distension of stomach
 - Intestinal-phase inhibitory stimuli
 - Distension of duodenum
 - Contents: osmolarity, acidity, fat

Gastric Motility

- Vomiting
 - Stimuli
 - Illness
 - Emotions
 - Pain
 - Distension in GI tract
 - Rotation of head
 - Ingestion of certain substances (emetics)
 - Reflex mediated through vomiting center in medulla

Motility of the Small Intestine

- Segmentation mixes chyme
- Peristalsis propels chyme forward
- Regulation of motility in small Intestine
- Distension: increases motility
- Autonomic nervous system
 - Parasympathetic → excites
 - Sympathetic → inhibits
- Hormones: gastrin stimulates motility

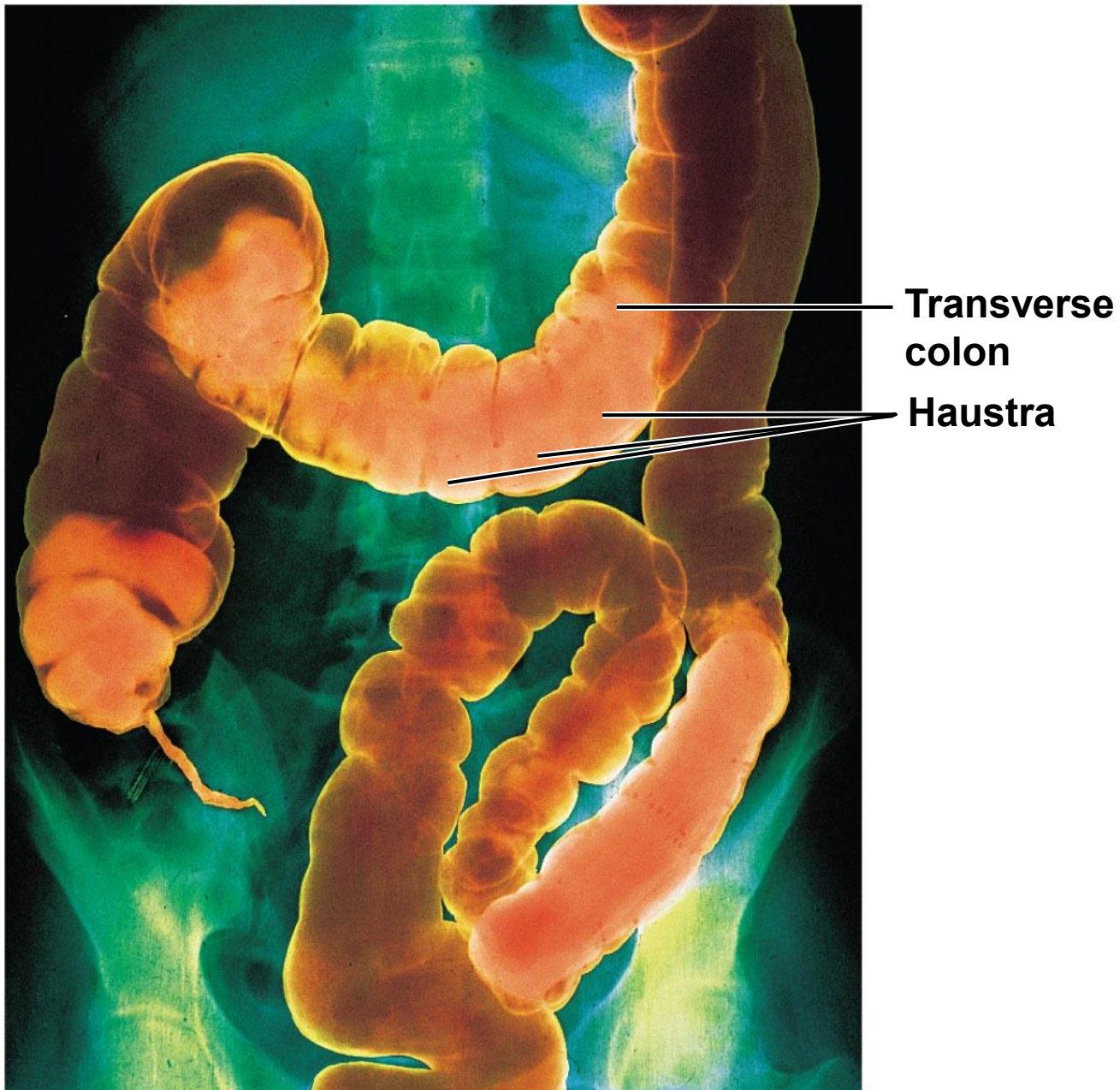
Motility of the Small Intestine

- Motility reflexes
 - Intestino-intestinal reflex
 - Injury or severe stress inhibits intestinal contractions
 - Ileogastric reflex
 - Distension of ileum inhibits gastric motility
 - Gastroileal reflex
 - Presence of chyme in stomach increases motility in ileum

Motility of the Colon

- Haustrations
 - Proximal colon
 - Like segmentation, but slower
- Mass movement
 - Propels bolus (material) toward rectum
 - Similar to peristalsis
 - Contraction lasts longer than relaxation

Figure 20.32 An X-ray image of a human colon showing haustra.



Motility of the Colon

- Regulation of colon motility
 - Colonocolonic reflex
 - Distension of colon in one area causes relaxation of other areas of colon
 - Gastrocolic reflex
 - Food in stomach increases colonic motility

Motility of the Colon

- Defecation
 - Voluntary and involuntary control
 - Distension of colon activates stretch receptors
 - Smooth muscle of rectum contracts
 - Increases pressure in rectum
 - Peristaltic contractions of sigmoid colon propel more fecal material into the rectum, increasing pressure
 - Internal anal sphincter relaxes simultaneously as external anal sphincter contracts
 - Both sphincters relax
 - Defecation